Classical training of conditional instantaneous quantum polynomial-time Born machines

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Instantaneous quantum polynomial-time (IQP) Born machines admit classical estimation of training observables for suitable input states, allowing model optimization without quantum circuit sampling. However, their standard fixed-input formulation models a single distribution. In this work, we introduce a conditional framework that encodes conditions into input states before applying a learned IQP unitary shared across conditions. The input preparation need not have strict IQP form, provided its reference state supports efficient classical sampling and amplitude-ratio evaluation. These operations enable unbiased classical estimation of Pauli-Z moments and their gradients. Experiments on sixteen-qubit discrete Gaussian families and the Curie–Weiss model demonstrate conditional generation. A phase-conditioned Gaussian input supports two-dimensional generation and generalization to mean–width combinations absent from fitting and development. For Curie–Weiss, the shared model achieves lower mean full-distribution total variation distance than both independent baseline families at two of three held-out temperatures, although the baselines are fitted directly at those temperatures and each family uses three times the total parameters and optimizer updates. Additional runs demonstrate the computational feasibility of classical conditional training from 64 to 2048 qubits. A sixteen-qubit signed-graph construction further combines correlated temperature encoding with exact switching symmetry and finite-data learning of two shared angles.

I. INTRODUCTION

Conditional generative models approximate a family of target distributions $p(\cdot \mid c)$ with a model $q_{\boldsymbol{\theta}}(\cdot \mid c)$, where c denotes the condition and $\boldsymbol{\theta}$ contains learned parameters shared across conditions. Classical examples include conditional generative adversarial networks [1–3], conditional variational autoencoders [4–6], and conditional diffusion models [7–9]. In the quantum domain, researchers have also adapted quantum circuit Born machines (QCBMs) [10, 11] to conditional tasks [12]. Researchers can introduce conditioning through input-state preparation or dedicated condition registers before applying a shared parameterized circuit [13–16]. Alternatively, postselection on a learned joint Born distribution can also yield conditional samples [17]. Beyond QCBMs, quantum diffusion models have explored conditional generation of both classical data and quantum states [18–20].

Existing complexity-theoretic results support the conjecture that classical algorithms cannot efficiently sample from certain output families of instantaneous quantum polynomial-time (IQP) Born machines under standard assumptions [21–25]. Yet, for maximum mean discrepancy (MMD) training with a Gaussian kernel [26], as commonly used for QCBMs [10, 27], classical algorithms can efficiently estimate the required Pauli-Z expectation values and their parameter gradients [28]. These estimators build on Van den Nest's techniques for computationally tractable input states [29]. However, the standard IQP formulation models a single target distribution per circuit. Extending IQP Born machines to conditional generation requires a conditioning mechanism that preserves classical access to the training observables and their gradients.

In this work, we develop a classically trainable framework for conditional IQP Born machines. We apply one learned IQP unitary $U_{\boldsymbol{\theta}}$ to condition-dependent input states $|\phi(c)\rangle$, giving $q_{\boldsymbol{\theta}}(x \mid c) = |\langle x|U_{\boldsymbol{\theta}}|\phi(c)\rangle|^2$. We establish sufficient conditions on the encoder for efficient classical estimation of the conditional training observables and their gradients, and construct encoders satisfying these conditions. Specifically, for the reference amplitudes $\psi_c(z) = \langle z|H^{\otimes n}|\phi(c)\rangle$, we require efficient sampling from $|\psi_c|^2$ and evaluation of amplitude ratios.

Building on sparse-observable methods [29, 30] and IQP training methods [28], we derive unbiased estimators of conditional Pauli-Z moments and their derivatives with respect to the learned phases. We bound the second moments of the moment and derivative estimators by one and four, respectively. These bounds imply that $O(\epsilon^{-2})$ reference samples suffice to estimate each specified moment or derivative to additive error ϵ at fixed confidence. We combine these estimators to obtain unbiased finite-sample estimates of the conditional MMD^2 objective and its gradients, enabling classical optimization of a single IQP Born machine across conditions. Figure 1 summa-

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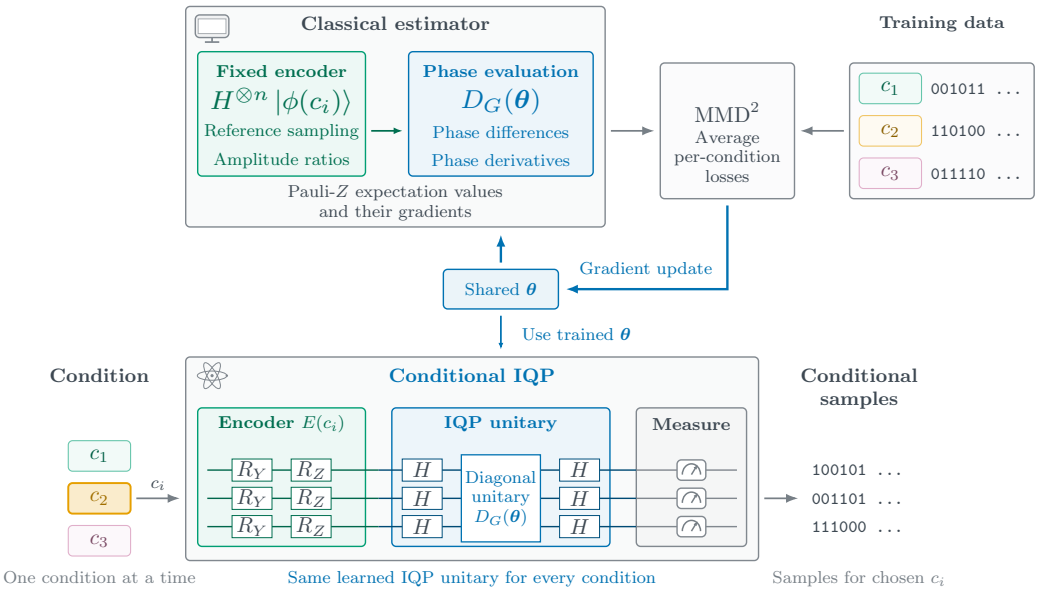


FIG. 1. Proposed framework for classical training and quantum sampling of conditional instantaneous quantum polynomial-time (IQP) Born machines. An encoder maps each condition c_i to an input state $|\phi(c_i)\rangle$. During classical training, samples and amplitude ratios from the reference state $H^{\otimes n}|\phi(c_i)\rangle$, together with phase differences induced by the IQP layer, yield estimates of conditional Pauli-Z moments and their gradients. We use these estimates to optimize one shared parameter vector $\boldsymbol{\theta}$ with a squared maximum mean discrepancy (MMD) objective averaged across training conditions. During quantum sampling, a chosen condition c_i determines the input state. Applying the trained IQP unitary $U_{\boldsymbol{\theta}}$ and measuring in the computational basis produces samples from the learned conditional distribution $q_{\boldsymbol{\theta}}(\cdot \mid c_i)$.

rizes the framework.

We evaluate the framework on discrete Gaussian families and Curie–Weiss distributions at sixteen qubits. A product-state encoder first illustrates conditional translation of Gaussian profiles. We then encode both mean and width through phases of a fixed correlated input and calibrate one angle shared across sixteen local rotations. Training the two coordinates separately tests generalization to unseen joint combinations, including pairs outside the joint training hull but within both marginal ranges.

For the Curie–Weiss model, we train a single model at four inverse temperatures. The model captures the change from unimodal to bimodal magnetization profiles, including the emergence of bimodality at the held-out inverse temperature $\beta = 1.0$. The encoder represents temperature through input phases while maintaining uniform computational-basis probabilities. The learned circuit converts this phase dependence into temperature-dependent output distributions. We compare this model with conditional and unconditional baseline families. Each family contains three independent circuits that we fit directly at the evaluation temperatures and uses three times the total parameters and optimizer updates of the shared model. The shared model achieves lower mean full-distribution total variation distance (TVD) than both baseline families at two of the three held-out temperatures, including $\beta = 1.6$, beyond the training range.

We also assess computational feasibility through optimization runs from 64 to 2048 qubits, tracking sampled MMD^2 at held-out temperatures throughout training.

Input-state design also allows exact output symmetries without restricting the learned IQP phases. A sixteen-qubit signed-graph experiment encodes temperature through correlated triangle phases and imposes sixty-four vertex switches at every condition. We train two shared local angles using 128 target samples at each of four temperatures and evaluate conditional graph order beyond the fitting range. Appendix C specifies the reference access, finite-data protocol and accuracy limitations.

The remainder of this paper is organized as follows. In Section II, we introduce the conditional IQP model and its classical estimators. In Section III, we develop the conditional MMD training protocol. In Section IV, we describe the numerical experiments and report their results. In Section V, we show how input-state preparation can enforce output symmetries. In Section VI, we discuss the implications and limitations of our findings, and in Section VII, we summarize the contributions.

II. CONDITIONAL IQP MODEL

We separate condition encoding from the learned unitary transformation. An encoder maps each condition

$c \in \mathcal{C}$ to an input state $|\phi(c)\rangle$, and we apply the same parameterized IQP unitary $U_{\boldsymbol{\theta}}$ to every encoded state. For a target family $\{p(\cdot \mid c) : c \in \mathcal{C}\}$ over n-bit strings, we define the model distribution as

$$q_{\boldsymbol{\theta}}(x \mid c) = \left|\langle x| U_{\boldsymbol{\theta}} |\phi(c)\rangle\right|^2, \qquad x \in \{0,1\}^n. \tag{1}$$

Throughout this work, we use "IQP unitary" to denote only the learned block $U_{\boldsymbol{\theta}}$. The encoder need not have IQP form and may include non-diagonal operations, so the complete conditional circuit need not belong to the strict IQP circuit class. Our conditions constrain classical access to the reference state rather than the encoder's gate structure: the reference state must support efficient sampling and amplitude-ratio evaluation.

The encoder also constrains how the model's output probabilities depend on the condition c. For rotation encodings with angles linear in c, the encoding generators determine the accessible Fourier frequencies, while the learned parameters control their coefficients within the constraints of the circuit architecture [31]. Multidimensional Fourier analyses further describe how the encoding and circuit architecture constrain the model's expressive capacity [32]. Caro et al. also derive encoding-dependent generalization bounds for parameterized quantum circuits [33]. We therefore distinguish encoder expressivity and generalization from the requirements for efficient classical estimation.

We parameterize the shared IQP unitary by an undirected graph $G = (V, E)$ with $n = |V|$ vertices. Each vertex $i \in V$ contributes a one-body generator Z_i, and each edge $\{i, j\} \in E$ contributes a two-body generator $Z_i Z_j$. We write the unitary as

$$U_{\boldsymbol{\theta}} = H^{\otimes n} D_G(\boldsymbol{\theta}) H^{\otimes n}, \tag{2}$$

where

$$D_G(\boldsymbol{\theta}) = \exp\left(i \sum_{i \in V} \theta_i Z_i + i \sum_{\{i,j\} \in E} \theta_{ij} Z_i Z_j\right). \tag{3}$$

All generators in D_G are diagonal in the computational basis and commute. In the numerical experiments, we use only one- and two-body generators. However, this formulation extends naturally to k-body interactions by adding the corresponding Pauli-Z products to the exponent.

We train the model using samples and amplitude ratios from the reference state $|\psi_c\rangle$ immediately before the diagonal IQP layer, without sampling the circuit's output distribution. We define its computational-basis amplitudes as

$$\psi_c(z) = \langle z| H^{\otimes n} |\phi(c)\rangle, \tag{4}$$

The squared amplitudes $|\psi_c(z)|^2$ give the reference sampling distribution for condition c. We require encoders that support two efficient classical operations for the relevant conditions c and bit masks a:

1. Sample $z \sim |\psi_c(z)|^2$.
2. Evaluate the amplitude ratio

$$R_c(z, a) = \frac{\psi_c(z \oplus a)}{\psi_c(z)} \tag{5}$$

for sampled z, where $\oplus$ denotes bitwise addition modulo two.

We call these two operations the sampling-and-ratio interface.

We estimate Pauli-Z moments by combining reference amplitude ratios with phase factors from the IQP layer. For each sampled string z, the ratio $R_c(z, a)$ compares the reference amplitudes of z and the paired string $z \oplus a$. The IQP layer contributes a complex phase factor determined by the phase difference between these strings. We multiply the amplitude ratio by this phase factor and average the real part over reference samples to estimate the corresponding Pauli-Z moment [29].

We evaluate amplitude ratios only on the support of $|\psi_c|^2$, because reference sampling never selects strings with $\psi_c(z) = 0$. Computationally tractable reference states in the sense of Van den Nest admit efficient computational-basis sampling and amplitude evaluation [29]. These states provide a sufficient realization of the sampling-and-ratio interface when amplitude evaluation supplies the precision needed to compute the required ratios efficiently.

The simplest case is a product-state encoder $|\phi(c)\rangle = \bigotimes_{i=1}^n |\phi_i(c)\rangle$. Its reference amplitudes factorize as $\psi_c(z) = \prod_i \psi_{c,i}(z_i)$, where $\psi_{c,i}(b) = \langle b| H |\phi_i(c)\rangle$. We can therefore generate reference samples by drawing each bit independently, and the amplitude ratio is

$$R_c(z, a) = \prod_{i : a_i = 1} \frac{\psi_{c,i}(1 - z_i)}{\psi_{c,i}(z_i)}. \tag{6}$$

When the local amplitudes have constant arithmetic evaluation cost, both operations require at most $O(n)$ local operations.

Efficient classical estimation of the observables used for training does not imply efficient classical sampling from $q_{\boldsymbol{\theta}}(\cdot \mid c)$. Existing IQP sampling-hardness results apply to specific circuit families and do not establish hardness for arbitrary condition-dependent input states or trained parameter values [22, 23]. To generate conditional samples on a quantum computer, we also need efficient circuits that prepare $|\phi(c)\rangle$ and implement $U_{\boldsymbol{\theta}}$. The classical sampling-and-ratio interface alone does not guarantee such circuits.

A. Conditional Pauli estimator and gradients

We express the conditional MMD objective in Section III in terms of Pauli-Z moments. For a bit mask $a \in \{0,1\}^n$, we define $Z_a = \bigotimes_{i=1}^n Z_i^{a_i}$ and denote its

conditional model expectation by $\mu_q(a \mid c)$. We use the sampling-and-ratio interface to derive estimators for this moment and its derivatives with respect to the learned phases.

We denote the diagonal IQP layer by $D_{\boldsymbol{\theta}}$ and write its action as

$$D_{\boldsymbol{\theta}} |z\rangle = e^{i f_{\boldsymbol{\theta}}(z)} |z\rangle , \tag{7}$$

with

$$f_{\boldsymbol{\theta}}(z) = \sum_{j=1}^{P} \theta_j \chi_{g_j}(z), \qquad \chi_{g_j}(z) = (-1)^{g_j \cdot z}, \tag{8}$$

where P is the number of generators and $g_j \in \{0,1\}^n$ specifies the support of the jth Pauli-Z generator. We evaluate the dot product $g_j \cdot z$ modulo two. We define the phase difference associated with the mask a as

$$\Delta_a f_{\boldsymbol{\theta}}(z) = f_{\boldsymbol{\theta}}(z \oplus a) - f_{\boldsymbol{\theta}}(z). \tag{9}$$

For a reference sample $z \sim |\psi_c|^2$, we define

$$\begin{aligned} W_{c,a}(z;\boldsymbol{\theta}) &= R_c(z,a) e^{i \Delta_a f_{\boldsymbol{\theta}}(z)}, \\ Y_{c,a}(z;\boldsymbol{\theta}) &= \mathrm{Re}\, W_{c,a}(z;\boldsymbol{\theta}). \end{aligned} \tag{10}$$

We define the derivative of the phase difference with respect to θ_j as

$$\begin{aligned} d_{j,a}(z) &= \partial_{\theta_j} \Delta_a f_{\boldsymbol{\theta}}(z) \\ &= \chi_{g_j}(z \oplus a) - \chi_{g_j}(z) \in \{-2, 0, 2\}. \end{aligned} \tag{11}$$

The following proposition establishes unbiased estimators for conditional Pauli-Z moments and their phase derivatives, and bounds their sampling error. We obtain these estimators by adapting amplitude-ratio methods for sparse observables [30]. For computationally tractable reference states, efficient estimation also follows from Van den Nest's sparse-operator theorem [29].

Proposition 1 (Classical estimation from reference-state access)**.** *Fix a condition c, a bit mask a, and a phase function with polynomially many diagonal Pauli-Z terms, as in Eq. (8). We assume that the reference state is normalized and independent of $\boldsymbol{\theta}$. We also assume that we can efficiently draw independent samples from $|\psi_c|^2$ and evaluate $R_c(z,a)$ for each sampled z. For a reference sample $z \sim |\psi_c|^2$, we define the moment and derivative estimators as follows:*

$$\begin{aligned} Y &= \mathrm{Re}\, W_{c,a}(z;\boldsymbol{\theta}), \\ Y'_j &= \mathrm{Re}[i\, d_{j,a}(z) W_{c,a}(z;\boldsymbol{\theta})] \end{aligned} \tag{12}$$

With exact sampling and arithmetic, these estimators have the following expectations and second-moment bounds:

$$\begin{aligned} \mathbb{E}[Y] &= \mu_q(a \mid c), & \mathbb{E}[Y^2] &\le 1, \\ \mathbb{E}[Y'_j] &= \partial_{\theta_j} \mu_q(a \mid c), & \mathbb{E}[(Y'_j)^2] &\le 4. \end{aligned} \tag{13}$$

We obtain unbiased estimates of the moment and its phase derivative by averaging Y and Y'_j over S independent reference samples. Their mean-square errors are at most $1/S$ and $4/S$, respectively. We can therefore estimate a specified moment or phase derivative to additive error ϵ at fixed confidence using $S = O(\epsilon^{-2})$ reference samples. These guarantees hold even when some reference amplitudes vanish.

Proof. We define the support of the reference distribution as $S_c = \{z : \psi_c(z) \neq 0\}$. Using $HZH = X$, we conjugate the measured observable Z_a through the final Hadamard layer to obtain $X_a = \bigotimes_{i=1}^n X_i^{a_i}$. This operator flips the bits selected by a, so

$$\begin{aligned} \mu_q(a \mid c) &= \langle \psi_c | D_{\boldsymbol{\theta}}^\dagger X_a D_{\boldsymbol{\theta}} |\psi_c\rangle \\ &= \mathrm{Re} \sum_{z \in S_c} \psi_c(z)^* \psi_c(z \oplus a) e^{i \Delta_a f_{\boldsymbol{\theta}}(z)} \\ &= \mathbb{E}_{z \sim |\psi_c|^2}[Y]. \end{aligned} \tag{14}$$

The last equality rewrites the sum as an expectation under $|\psi_c|^2$, using the amplitude ratio and the definition of Y.

The reference state does not depend on $\boldsymbol{\theta}$, so its sampling distribution and amplitude ratios remain fixed. We therefore differentiate only the IQP phase factor in the finite sum to obtain

$$\begin{aligned} \partial_{\theta_j} \mu_q(a \mid c) &= \mathbb{E}_{z \sim |\psi_c|^2}[\mathrm{Re}\,(i\, d_{j,a}(z) W_{c,a}(z;\boldsymbol{\theta}))] \\ &= \mathbb{E}[Y'_j]. \end{aligned} \tag{15}$$

Thus Y'_j is an unbiased estimator of the phase derivative.

Although individual amplitude ratios can be large, normalization bounds their second moment. The phase factor has unit magnitude, and the sampling probability $|\psi_c(z)|^2$ cancels the denominator in $|R_c(z,a)|^2$ when we take the expectation. We therefore obtain

$$\mathbb{E}[|W_{c,a}|^2] = \sum_{z \in S_c} |\psi_c(z \oplus a)|^2 \le 1. \tag{16}$$

The final inequality follows because bit flipping maps S_c to a subset of all bit strings and the reference state is normalized. Since Y is the real part of $W_{c,a}$, we have $\mathbb{E}[Y^2] \le 1$. Combining $|d_{j,a}(z)| \le 2$ with the same bound gives $\mathbb{E}[(Y'_j)^2] \le 4$. Averaging Y and Y'_j over S independent reference samples therefore gives mean-square errors at most $1/S$ and $4/S$, respectively. Chebyshev's inequality then yields the stated fixed-confidence sample complexity. The argument allows vanishing reference amplitudes because we evaluate ratios only for sampled strings. □

In practice, we draw S independent reference samples $z_s \sim |\psi_c|^2$ and estimate the moment as

$$\widehat{\mu}_q(a \mid c) = \frac{1}{S} \sum_{s=1}^{S} Y_{c,a}(z_s;\boldsymbol{\theta}). \tag{17}$$

We estimate the phase derivative by averaging the corresponding Y'_j values over the same samples. For the standard IQP input $|0\rangle^{\otimes n}$, all reference amplitudes are equal, so $R_c(z,a) = 1$. The moment estimator then reduces to $Y = \cos[\Delta_a f_{\boldsymbol{\theta}}(z)]$, recovering the usual IQP cosine estimator [29, 34].

For a circuit with r ancillary qubits, we apply the same estimator to the joint reference state, provided it supports the sampling-and-ratio interface. If we discard the ancillary measurement outcomes, we extend each visible-bit mask a to $(a, 0^r)$, where 0^r denotes r zeros. The zero entries ensure that the observable acts as the identity on the ancillary qubits.

III. CONDITIONAL IQP TRAINING

We now formulate the training protocol for conditional IQP Born machines. Throughout this work, we use *classically trainable* to mean that we can efficiently estimate the training objective and its phase gradients to a prescribed additive accuracy on a classical computer.

A. Condition-balanced MMD objective

We optimize the shared parameter vector $\boldsymbol{\theta}$ by minimizing the squared maximum mean discrepancy (MMD) across the training conditions. For distributions p and q on $\{0,1\}^n$ and a positive-definite kernel k, the population squared MMD is [26]

$$\begin{aligned}\mathrm{MMD}^2(p,q) = \mathbb{E}_{x,x'\sim p}[k(x,x')] - 2\mathbb{E}_{x\sim p,\, y\sim q}[k(x,y)] \\ + \mathbb{E}_{y,y'\sim q}[k(y,y')],\end{aligned} \tag{18}$$

where we draw the two samples in each expectation independently.

Throughout this work, we use the Gaussian kernel on binary strings,

$$k_\sigma(x,y) = \exp\left[-\frac{d_H(x,y)}{2\sigma^2}\right], \tag{19}$$

where $d_H(x,y)$ is the Hamming distance, which counts the positions at which x and y differ, and $\sigma > 0$ is the kernel bandwidth.

Let $\mathcal{C}_{\mathrm{train}}$ denote the set of training conditions and τ the set of kernel bandwidths. We train the shared model by minimizing the condition-balanced objective

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{C}_{\mathrm{train}}|}\sum_{c\in\mathcal{C}_{\mathrm{train}}}\frac{1}{|\tau|}\sum_{\sigma\in\tau}\mathrm{MMD}^2_\sigma(p(\cdot\mid c), q_{\boldsymbol{\theta}}(\cdot\mid c))\,. \tag{20}$$

Here, MMD^2_σ denotes the squared MMD computed with the kernel k_σ. We weight all training conditions and bandwidths equally. At each optimizer update, we independently draw one condition uniformly from $\mathcal{C}_{\mathrm{train}}$ and one bandwidth uniformly from τ, giving a stochastic estimate of this average.

B. Character expansion

For binary strings, the Gaussian kernel admits an exact expansion in binary characters. We sample a visible-bit mask $a \in \{0,1\}^n$ by drawing its coordinates independently according to

$$a_i \sim \mathrm{Bernoulli}(p_\sigma), \qquad p_\sigma = \frac{1 - e^{-1/(2\sigma^2)}}{2}, \tag{21}$$

The corresponding binary character is

$$\chi_a(x) = (-1)^{a\cdot x}, \tag{22}$$

which records the parity of the bits selected by a. Averaging the product of these characters over a gives

$$k_\sigma(x,y) = \mathbb{E}_a[\chi_a(x)\chi_a(y)]\,. \tag{23}$$

For a condition c, define the target and model character moments

$$\begin{aligned}\mu_p(a\mid c) &= \mathbb{E}_{x\sim p(\cdot|c)}[\chi_a(x)],\\ \mu_q(a\mid c) &= \mathbb{E}_{x\sim q_{\boldsymbol{\theta}}(\cdot|c)}[\chi_a(x)].\end{aligned} \tag{24}$$

Substituting Eq. (23) into Eq. (18) gives

$$\mathrm{MMD}^2_\sigma(p(\cdot\mid c), q_{\boldsymbol{\theta}}(\cdot\mid c)) = \mathbb{E}_a\left[\left(\mu_p(a\mid c) - \mu_q(a\mid c)\right)^2\right]. \tag{25}$$

Thus, the squared MMD averages the squared differences between target and model character moments over the mask distribution.

For a Born machine, the visible Pauli operator

$$Z_a = \bigotimes_{i=1}^n Z_i^{a_i} \tag{26}$$

has eigenvalue $\chi_a(x)$ on the computational-basis state $|x\rangle$. Its model expectation value is therefore

$$\mu_q(a\mid c) = \langle Z_a\rangle_{q_{\boldsymbol{\theta}}(\cdot|c)}. \tag{27}$$

We can thus express the conditional squared MMD entirely in terms of visible Pauli-Z moments. We estimate the model moments classically using the reference-state estimator in Sec. II A.

The bandwidth σ determines which correlation orders the objective emphasizes. The mask's Hamming weight $K = |a|$ counts the bits it selects, so it gives the correlation order of Z_a. Because we sample the mask coordinates independently,

$$\begin{aligned}K &\sim \mathrm{Binomial}(n, p_\sigma),\\ \mathbb{E}[K] &= np_\sigma,\\ \mathrm{Var}(K) &= np_\sigma(1-p_\sigma).\end{aligned} \tag{28}$$

Each bandwidth therefore induces a distribution over correlation orders. Large bandwidths favor lower-order

correlations, whereas decreasing σ shifts weight toward higher-order correlations, with $\mathbb{E}[K] \to n/2$ as $\sigma \to 0$.

The choice of bandwidth can also influence the model's trainability. Previous results demonstrate that cost functions consisting of global terms can induce barren plateaus [35, 36], even in shallow quantum circuits [37]. This creates a trade-off between trainability and sensitivity to higher-order correlations. Improving trainability may require restricting the objective to low-body correlations, which limits its ability to distinguish distributions that differ in higher-order correlations [38]. Researchers have proposed several strategies to address these difficulties, including tailored generator sets and restricted initializations [39, 40], and adversarial kernel adaptation [41]. How best to balance trainability and sensitivity to higher-order correlations remains an open question beyond the scope of this work.

C. Unbiased finite-sample MMD and gradients

For a fixed condition c and visible mask a, we draw $B \geq 2$ independent target samples and compute

$$T_b = \chi_a(x_b), \qquad x_b \sim p(\cdot \mid c). \tag{29}$$

Independently of these target samples, we draw $S \geq 2$ independent reference samples and evaluate the reference-state estimator from Eq. (10),

$$Y_s = Y_{c,a}(z_s; \boldsymbol{\theta}), \qquad z_s \sim |\psi_c|^2. \tag{30}$$

We denote the corresponding sample means by

$$\overline{T} = \frac{1}{B}\sum_{b=1}^{B} T_b, \qquad \overline{Y} = \frac{1}{S}\sum_{s=1}^{S} Y_s. \tag{31}$$

The direct estimate $(\overline{Y} - \overline{T})^2$ overestimates the squared moment difference on average because it includes the sampling variances of both sample means. We remove this bias by excluding products that use the same draw twice:

$$\begin{aligned}\widehat{\Delta_c^2}(a) = &\frac{1}{S(S-1)}\sum_{s\neq t} Y_s Y_t - 2\overline{Y}\,\overline{T} \\ &+ \frac{1}{B(B-1)}\sum_{b\neq b'} T_b T_{b'}.\end{aligned} \tag{32}$$

Independence within each pair gives

$$\mathbb{E}\left[\widehat{\Delta_c^2}(a)\right] = \left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)^2. \tag{33}$$

Thus, Eq. (32) provides an unbiased finite-sample estimator of the squared character-moment difference appearing in Eq. (25). An unbiased finite-sample estimate of MMD^2 may be negative.

We evaluate the pair sums using the identities

$$\sum_{s\neq t} Y_s Y_t = \left(\sum_s Y_s\right)^2 - \sum_s Y_s^2, \tag{34}$$

and, since $T_b^2 = 1$,

$$\sum_{b\neq b'} T_b T_{b'} = \left(\sum_b T_b\right)^2 - B. \tag{35}$$

Once we have computed the sample values, we can therefore evaluate the estimator in time linear in the numbers of target and reference samples.

The same construction gives an unbiased estimator of the phase gradients. For a phase parameter θ_j, let

$$Y_s' = \partial_{\theta_j} Y_{c,a}(z_s; \boldsymbol{\theta}). \tag{36}$$

We evaluate Y_s' using Eq. (15). Differentiating Eq. (32) gives

$$\partial_{\theta_j}\widehat{\Delta_c^2}(a) = \frac{2}{S(S-1)}\sum_{s\neq t} Y_s Y_t' - 2\overline{T}\,\overline{Y'}, \tag{37}$$

where

$$\overline{Y'} = \frac{1}{S}\sum_s Y_s'. \tag{38}$$

Taking the expectation yields

$$\mathbb{E}\left[\partial_{\theta_j}\widehat{\Delta_c^2}(a)\right] = 2\left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)\partial_{\theta_j}\mu_q(a \mid c), \tag{39}$$

which is the gradient of the corresponding squared moment difference.

At a fixed parameter vector, the expectations above average over independent draws from the target distribution and the reference sampling law. During training, we sample target minibatches with replacement from fixed training pools; the stochastic gradient is therefore unbiased for the empirical objective associated with those pools.

We exclude the diagonal terms $s = t$ because Y_s and Y_s' can be correlated when we compute them from the same reference draw. The fixed encoder makes the reference sampling law independent of $\boldsymbol{\theta}$. We therefore hold the sampled reference strings fixed during automatic differentiation to evaluate the phase gradients without differentiating through the sampler.

D. Stochastic mask estimator

Equation (25) expresses the squared MMD as an expectation over the mask distribution in Eq. (21). Since there are 2^n possible masks, we estimate this expectation using a finite sample of masks.

The identity mask $a = 0$ satisfies

$$\mu_p(0 \mid c) = \mu_q(0 \mid c) = 1 \tag{40}$$

for every condition and therefore contributes zero to the squared MMD. We consequently sample only nonidentity masks from the conditional distribution

$$P_\sigma(a \mid a \neq 0). \tag{41}$$

Under the original mask distribution, nonidentity masks have total probability

$$Q_\sigma^+ = 1 - (1 - p_\sigma)^n. \tag{42}$$

We multiply the conditional expectation by Q_σ^+ to preserve the original Gaussian kernel objective.

We draw M nonidentity masks $a_1, \ldots, a_M$ independently from $P_\sigma(a \mid a \neq 0)$ and estimate the squared MMD at condition c as

$$\widehat{\mathrm{MMD}}_\sigma^2(c) = Q_\sigma^+ \frac{1}{M} \sum_{m=1}^{M} \widehat{\Delta}_c^2(a_m), \tag{43}$$

where $\widehat{\Delta}_c^2$ is the unbiased finite-sample estimator in Eq. (32). Averaging over the mask, target, and reference draws recovers the full kernel squared MMD.

We evaluate each distinct sampled mask only once. Let $\mathcal{U}$ denote the set of distinct masks and m_u the number of times mask u occurs. We rewrite Eq. (43) as

$$\widehat{\mathrm{MMD}}_\sigma^2(c) = Q_\sigma^+ \sum_{u \in \mathcal{U}} \frac{m_u}{M} \widehat{\Delta}_c^2(u). \tag{44}$$

Weighting each distinct mask by m_u/M preserves the sampled objective and its gradient while reducing the number of Pauli moments we evaluate.

IV. NUMERICAL EXPERIMENTS

We evaluate the proposed architecture on Gaussian families controlled by translation or by both mean and width, and on the Curie–Weiss model at sixteen qubits. We also assess computational feasibility through optimization runs from 64 to 2048 qubits. For the sixteen-qubit benchmarks, we use the exact full-distribution total variation distance as the primary accuracy metric,

$$\mathrm{TVD}(p, q) = \frac{1}{2} \sum_{x \in \{0,1\}^n} |p(x) - q(x)|. \tag{45}$$

We also report the exact TVD between Hamming-weight profiles as a diagnostic,

$$\mathrm{TVD}_K(p, q) = \frac{1}{2} \sum_{k=0}^{n} |\rho_p(k) - \rho_q(k)|, \tag{46}$$

where $\rho_p(k)$ and $\rho_q(k)$ give the probabilities of observing Hamming weight k under p and q, respectively. Grouping bitstrings by Hamming weight gives $\mathrm{TVD}_K(p, q) \leq \mathrm{TVD}(p, q)$. The targets assign equal probability to all bitstrings with the same Hamming weight, but the learned models need not preserve this property. Profile TVD therefore does not resolve errors within Hamming-weight shells. We compute both metrics using the same exact circuit probabilities.

We train the models by minimizing the squared-MMD objective. We select checkpoints for the one-dimensional Gaussian translation experiment using mean exact profile TVD over the development conditions. The initial two-dimensional Gaussian benchmark and the trained sixteen-qubit Curie–Weiss models use exact full-distribution TVD for hyperparameter and checkpoint selection. We additionally evaluate the two-dimensional Gaussian model with finite-target validation, using empirical MMD for checkpoint selection. Appendices A and B specify the protocols and distinguish the target information available in each experiment.

A. Discrete Gaussian translation

Our first experiment illustrates the conditioning mechanism with a translated discrete Gaussian Hamming-weight profile. For $n = 16$ bits, let $K = |x|$ denote the Hamming weight of bitstring x. The condition $c \in [0, 1]$ controls the nominal center $5 + 6c$ of the profile. The target distribution assigns equal probability to all bitstrings with the same Hamming weight.

We use a fixed product-state encoder that applies the same two rotations to every qubit:

$$|\phi(c)\rangle = \bigotimes_{i=1}^{16} R_Z\left(c - \frac{1}{2}\right) R_Y(1.5)\, |0\rangle_i \,. \tag{47}$$

Changing c modifies the input phases but leaves the encoder's computational-basis measurement distribution unchanged.

The shared unitary uses a 4×4 lattice with 16 one-body phase gates and 24 two-body phase gates between horizontal and vertical nearest neighbors, giving 40 trainable parameters.

We train at $c \in \{0, 0.25, 0.5, 0.75, 1\}$ and evaluate at 80 held-out conditions. The median across runs of the mean full-distribution TVD over these conditions is 0.0995, compared with 0.2922 for the encoder alone (Table I). The corresponding profile TVD is 0.0464. The higher full-distribution TVD reveals discrepancies within Hamming-weight shells that the profile metric does not resolve. Figure 2 shows the translated profiles and conditional response. Appendix A 2 gives the training and evaluation protocol.

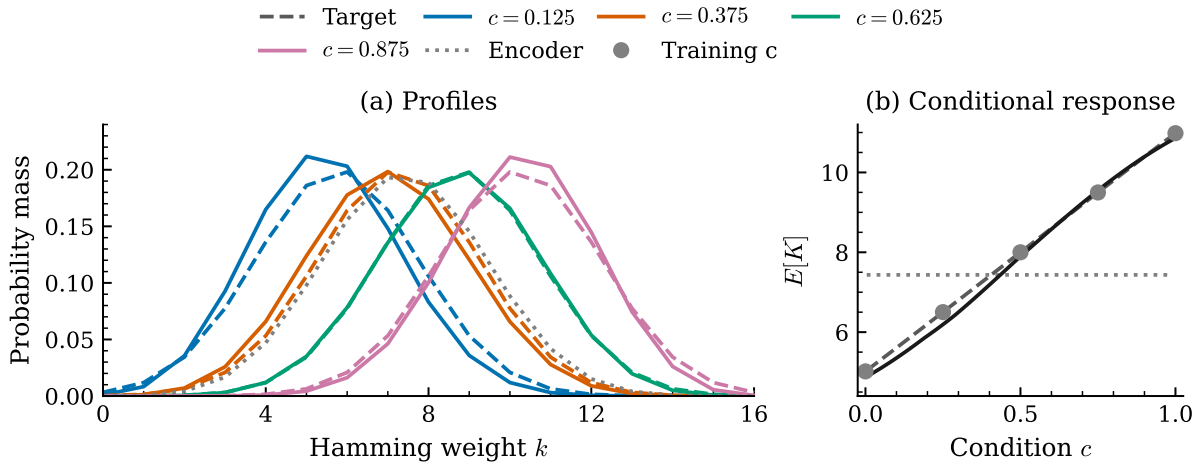


FIG. 2. Conditional translation of discrete Gaussian Hamming-weight profiles at sixteen qubits. (a) Solid colored curves show exact model profiles averaged across five runs at the development conditions $c \in \{0.125, 0.375, 0.625, 0.875\}$. We use these conditions for checkpoint selection and exclude them from fitting. (b) The solid black curve shows the median across runs of the model's mean output Hamming weight at each of 80 held-out conditions. Shading shows the interquartile range. Dashed curves show exact targets, dotted gray curves show the encoder alone, and gray circles mark training conditions on the target response. Table I reports exact full-distribution and profile errors.

TABLE I. Exact full-distribution and profile TVDs for the Gaussian benchmark. Each entry reports the median across five runs of the mean error over 80 held-out conditions. Bold entries mark the lowest error in each column.

Model	Full TVD ↓	Profile TVD ↓
Conditional IQP	**0.0995**	**0.0464**
Encoder only	0.2922	0.2922

B. Two-dimensional Gaussian conditioning and generalization

We next ask whether a shared model can generate distributions controlled by two coordinates and generalize to combinations absent from fitting. We extend Gaussian translation to changes in both center and width, while keeping the encoder amplitudes independent of the conditions. For sixteen bits, we define the target distribution by its Hamming-weight profile,

$$p(x \mid u, v) = \frac{\rho_{u,v}(|x|)}{\binom{16}{|x|}}, \tag{48}$$

$$\rho_{u,v}(k) \propto \exp\left[-\frac{(k-\mu(u))^2}{2\sigma_{\mathrm{G}}(v)^2}\right], \tag{49}$$

where $u, v \in [-1, 1]$, $\mu(u) = 8 + 3u$ and $\sigma_{\mathrm{G}}(v) = 2(1 + 0.2v)$. We normalize over $k \in \{0, \ldots, 16\}$. The coordinates control the nominal center and width; discretization and truncation cause small differences from the actual target moments.

We encode both coordinates through linear and quadratic phases of Hamming weight. These provide smooth conditional features without preparing the requested conditional profile in the input amplitudes. We first prepare one fixed state,

$$|G_0\rangle = \sum_{k=0}^{16} \sqrt{r_0(k)}\, |D_k^{(16)}\rangle\,, \qquad r_0(k) \propto e^{-(k-8)^2/8}, \tag{50}$$

where $|D_k^{(16)}\rangle$ is the normalized uniform superposition of bitstrings of weight k, and r_0 is normalized over the same finite support. With $J_z = \frac{1}{2}\sum_j Z_j$, we encode the conditions as

$$|\phi(u, v)\rangle = e^{-i(u/2)J_z} e^{-i(v/40)J_z^2} |G_0\rangle\,. \tag{51}$$

The first phase is linear in Hamming weight and consists of identical $R_Z(u/2)$ rotations. The second is quadratic in Hamming weight and introduces a collective phase controlled by v. Measuring the encoder alone always gives r_0, independently of either condition. Conditional changes in the output therefore arise when the subsequent rotation converts these phases into probabilities.

We use one learned angle shared across all sixteen local rotations,

$$U_\theta = e^{i\theta \sum_j X_j} = R_X(-2\theta)^{\otimes 16}. \tag{52}$$

The encoder and learned layer preserve permutation symmetry, so their bitstring probabilities are uniform within each Hamming-weight shell. Full-distribution TVD therefore equals profile TVD for this model. The reference state $H^{\otimes 16} |\phi(u, v)\rangle$ also has only seventeen distinct amplitudes, enabling efficient sampling and amplitude-ratio evaluation. Appendix B gives the reference operations and preparation costs.

We compare two training designs, each with nine fitting conditions and a fixed pool of 1024 target strings per condition. Separate variation changes u at $v = 0$ or v at $u = 0$, excluding mixed pairs from both fitting and development. Joint variation fits the 3×3 grid $\{-1, 0, 1\}^2$. The initial benchmark selects configurations and checkpoints using exact development TVD and evaluates five final fits per design. This provides a reference for conditional fidelity, but selection accesses target-distribution information beyond the finite fitting pools.

Separate variation gives mean reporting TVD 0.0258 ± 0.0005 over sixteen unseen mixed pairs. Four pairs interpolate within its joint fitting hull and give 0.0159 ± 0.0016; twelve lie outside that hull and give 0.0290 ± 0.0005. Every pair combines coordinate values encountered separately during fitting. Joint variation gives 0.0222 ± 0.0013 over thirty-two withheld pairs. We average errors over conditions within each fit, then report the mean and sample standard deviation across fits. Figure 3 shows the broader conditional domain. Response sweeps recover approximately 96–97% of the target mean displacement and 91–95% of the width change, with increasing width along every tested slice (Appendix B).

We then test whether this behavior persists when checkpoint selection uses finite target observations. With the encoder and optimizer fixed, five additional fits per design select checkpoints using empirical validation MMD and exact model moments. Each fit uses 9216 fitting samples and an additional 1024 validation samples for separate variation or 2560 for joint variation. Mean reporting TVD is 0.0288 ± 0.0058 and 0.0280 ± 0.0035, respectively. These fits recover approximately the full target mean displacement and 89–93% of the endpoint width change on the tested slices, without adjacent width reversals. Thus useful two-coordinate response survives finite-target validation. This check removes population target information from checkpoint selection, while retaining the earlier choices of encoder and optimizer.

Simple calibration controls clarify what fitting contributes. The model's analytic mean response allows us to estimate its shared angle directly from the finite fitting-pool means. This gives reporting TVD 0.0263 ± 0.0025 for separate variation and 0.0259 ± 0.0019 for joint variation, without iterative optimization or validation data. Replacing the fixed Gaussian prior with $|+\rangle^{\otimes 16}$ and calibrating the angle in the same way gives slightly lower errors, 0.0241 ± 0.0026 and 0.0230 ± 0.0019. The quadratic conditional phase remains entangling in this control. Appendix B gives the response identity, calibration rule, terminal-checkpoint results and estimator comparison.

Together, these experiments demonstrate structured conditional generation and generalization to mean–width combinations withheld from fitting and development. The conditions change only phases of a fixed input, and the readout converts those phases into distributional changes. The chosen features nevertheless supply a task-informed prior for both responses; fitting calibrates that prior rather than discovering an unknown conditional law. The calibration controls show that neither iterative optimization nor Gaussian amplitude preparation is necessary for the observed fidelity. Earlier studies informed the encoder and examined the reporting conditions, so we interpret the results as a conditional-capability demonstration rather than an independent confirmatory test.

C. Curie-Weiss spin model

We use the zero-field Curie–Weiss model [42] to test whether the architecture captures changes in distributional shape. The spins $s_i \in \{-1, +1\}$ interact with equal strength across all pairs, so the target assigns equal probability to bitstrings with the same Hamming weight. As the inverse temperature β increases, the finite-size Hamming-weight profile changes from a central peak to two symmetric modes.

We encode the inverse temperature using the fixed product-state encoder

$$|\phi(\beta)\rangle = \bigotimes_{i=1}^{16} R_Z(0.75\beta) R_Y\left(\frac{\pi}{2}\right) |0\rangle_i . \tag{53}$$

The R_Z rotations encode β in the input phases while leaving the computational-basis measurement distribution uniform. We parameterize the IQP unitary with all one- and two-body generators, giving 136 trainable parameters. We train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$ and evaluate at the held-out inverse temperatures $\beta \in \{0.4, 1.0, 1.6\}$.

We compare the shared conditional IQP model with two independently tuned baseline families. Each family comprises three circuits, which we fit separately at the three evaluation temperatures. The independent conditional IQPs retain the corresponding temperature encoder, whereas the independent unconditional IQPs use $|0\rangle^{\otimes 16}$ as their input state.

We use exact full-distribution TVD for hyperparameter optimization (HPO) and model selection. We tune the shared model on development temperatures and each independent family jointly across its fitted temperatures. Appendix A 3 a details the protocol.

Each independent baseline family uses three circuits and therefore three times the total parameters and optimizer updates of the shared model. The shared model uses more training-pool samples, with 4,096 entries per run compared with 3,072 for each independent family. Appendix A 3 reports the detailed budgets.

The shared conditional model captures the change from unimodal to bimodal magnetization profiles, including bimodality at the held-out inverse temperature $\beta = 1.0$ (Figure 4). It achieves lower mean full-distribution TVD than both independent baseline families at $\beta = 1.0$ and 1.6, with mean errors of 0.2175 and 0.1709, respectively (Table III). The temperature $\beta = 1.6$ lies beyond the training range. At $\beta = 0.4$, the independent conditional family achieves lower mean TVD than the shared

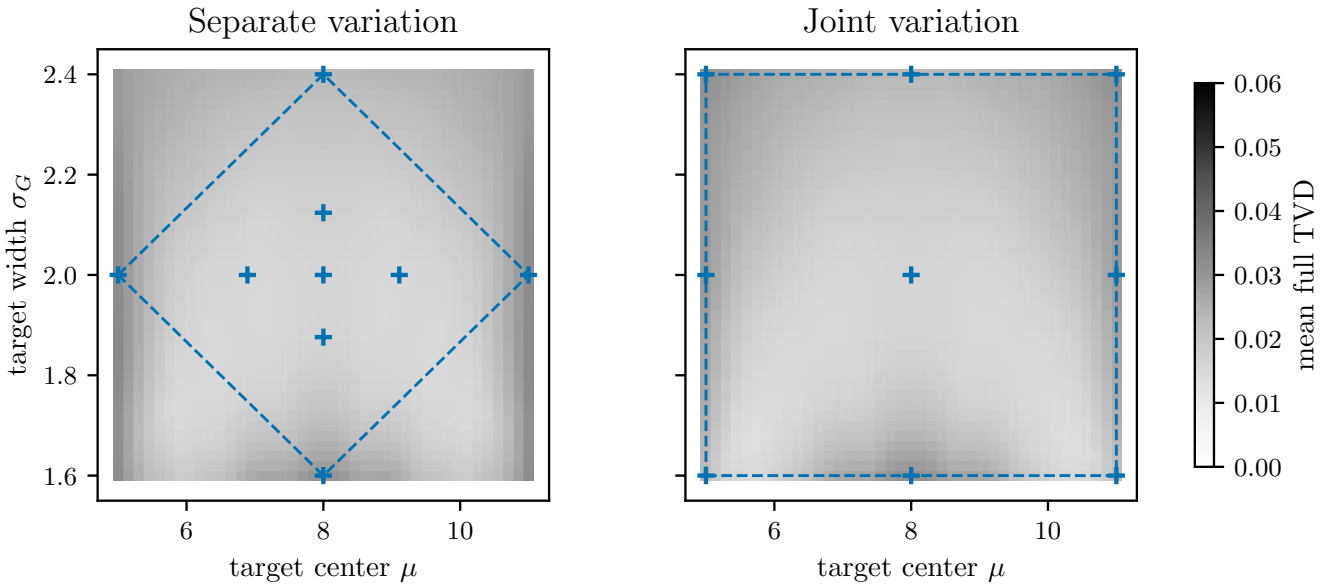


FIG. 3. Two-coordinate Gaussian generation with fixed input amplitudes and condition-dependent phases. Shading shows exact full-distribution TVD averaged across five fits selected with exact development TVD at each point of a 41×41 grid. Crosses mark fitting conditions and dashed lines bound their convex hulls. (a) Separate variation excludes mixed pairs from fitting and development. Points outside the diamond combine individually observed coordinate values beyond the joint fitting hull. (b) Joint variation fits a 3×3 grid and tests interpolation between its points. Both panels use the same color scale; the axes show nominal target center and width. We evaluate the grid after selection, including fitting and development points. Table II uses separate reporting sets that exclude fitting and development. Earlier exploratory studies informed the encoder and examined some reporting conditions.

TABLE II. Full-distribution TVD on each design's reporting set, excluding fitting and development conditions. For the sampled fits, we report the mean and sample standard deviation of per-fit mean errors across five runs. The fixed rotation was chosen after the capacity pilot and frozen before sampled training. Exact development TVD selects checkpoints. The designs use different reporting sets.

	Separate variation 16 reporting pairs	Joint variation 32 reporting pairs
Encoder only	0.3901	0.3612
Fixed $\theta = 3\pi/8$	0.0347	0.0309
Sampled calibration, selected checkpoint	0.0258 ± 0.0005	0.0222 ± 0.0013

TABLE III. Exact full-distribution TVD for sixteen-qubit Curie–Weiss models. For trained models, we report means ± sample standard deviations across five confirmation runs used for model selection and reporting. The encoder-only baseline is deterministic. We evaluate the shared model at temperatures excluded from fitting and model selection, whereas we fit the independent circuits directly at each evaluation temperature. Bold entries identify the lowest mean TVD in each column, including the encoder-only baseline.

Model	Evaluation	Full-distribution TVD ↓		
		$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Encoder only	Baseline	**0.1180**	0.4406	0.8699
Shared conditional IQP	Held-out	0.2240 ± 0.0115	**0.2175 ± 0.0136**	**0.1709 ± 0.0136**
Independent conditional IQPs	Fitted	0.1240 ± 0.0083	0.2309 ± 0.0319	0.3948 ± 0.0003
Independent unconditional IQPs	Fitted	0.3842 ± 0.0017	0.3322 ± 0.0105	0.3299 ± 0.0119

model (0.1240 versus 0.2240), while the uniform encoder-only baseline achieves the lowest error (0.1180). Together with the held-out results, this comparison suggests that learning across conditions may provide a useful inductive bias for generalization.

The profile diagnostic in Appendix A 3 (Table VIII)

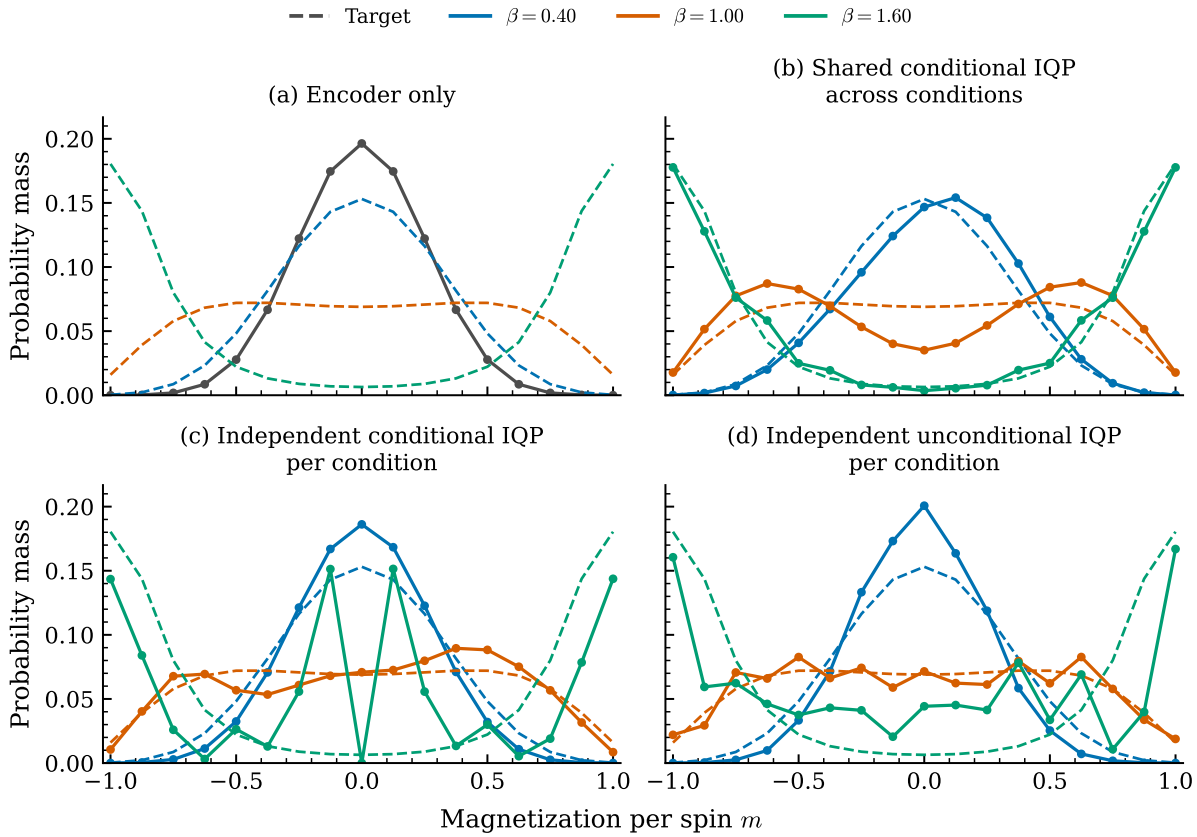


FIG. 4. Curie–Weiss magnetization profiles at sixteen qubits. We write the magnetization as $m = (2k - 16)/16$, where k is the Hamming weight. Panels show (a) encoder only, (b) shared conditional IQP, (c) independent conditional IQPs, and (d) independent unconditional IQPs. Colors distinguish $\beta = 0.4, 1.0, 1.6$, and dashed curves show exact targets. Solid curves show exact model profiles averaged over five confirmation runs used for model selection and reporting; the encoder-only profile is deterministic. We evaluate the shared model at temperatures excluded from fitting and model selection, whereas we fit the independent circuits directly at each displayed temperature. Table III reports full-distribution TVD.

gives a different ranking from full-distribution TVD. At $\beta = 1.0$, both independent baseline families achieve lower mean profile TVD than the shared model but higher mean full-distribution TVD. At $\beta = 0.4$, the shared model reduces mean profile TVD relative to the encoder while increasing mean full-distribution TVD. These differences show that agreement in Hamming-weight profiles does not ensure accurate bitstring probabilities.

We also evaluate physical observables to distinguish magnetization magnitude from energy fluctuations. The exact zero-field target has $\langle m \rangle = 0$ at finite size, so we report the mean absolute magnetization $\langle |m| \rangle$, with $m = n^{-1} \sum_i s_i$ and $n = 16$. For the Hamiltonian $H = -nm^2/2$, we define the per-spin heat-capacity fluctuation estimate

$$c_{\mathrm{fluc}}(\beta) = \frac{\beta^2}{n} \left(\langle H^2 \rangle - \langle H \rangle^2 \right). \tag{54}$$

This expression equals the per-spin equilibrium heat capacity for the Gibbs target. For a learned distribution, it measures energy fluctuations and need not equal the temperature derivative of the mean energy per spin. We compute both observables from exact probabilities for the same five selected confirmation circuits on a grid from $\beta = 0$ to 2 in steps of 0.02, without additional fitting or model selection. The sweep includes fitting and development temperatures as well as temperatures beyond the fitting range; it is not an entirely held-out evaluation. Appendix A 3 b specifies the calculation and aggregation.

The dense response reveals errors that the three-temperature distribution comparison does not fully describe (Figure 5). At $\beta = 1.0$, the shared model overestimates mean absolute magnetization (0.525 versus the target 0.451), while the independent fitted baselines are closer. At higher inverse temperatures, the shared response levels off and underestimates the target (0.855 versus 0.936 at $\beta = 2$). Energy fluctuations show a different discrepancy. The target's heat capacity reaches its maximum on the evaluated grid near $\beta = 1.52$ and then decreases, whereas the shared model's fluctuation estimate continues increasing through $\beta = 2$, reaching 1.299 versus the target 0.504. At $\beta = 1.6$, however, the shared estimate remains closer to the target than either indepen-

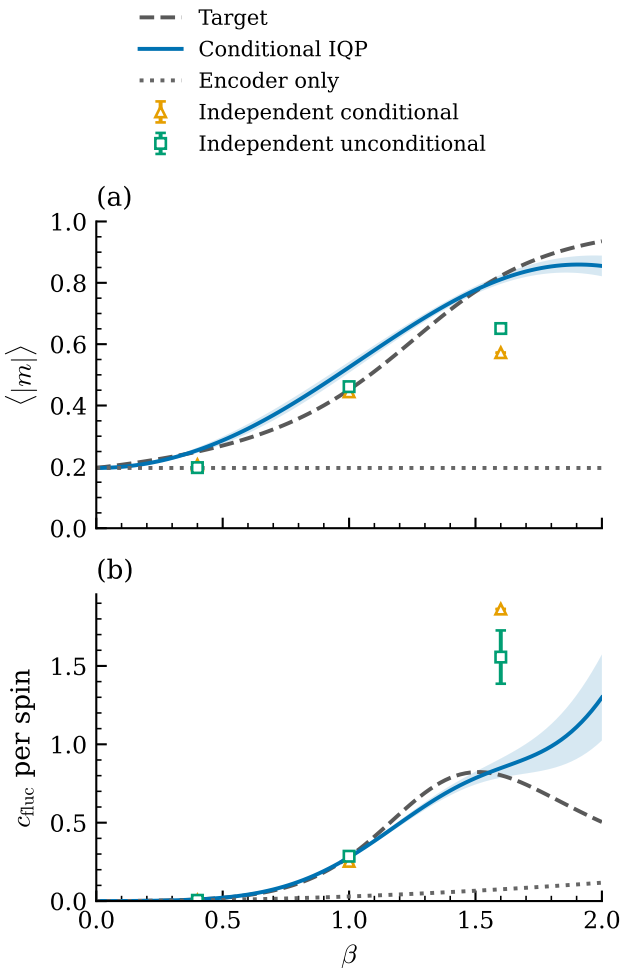


FIG. 5. Sixteen-qubit Curie–Weiss observables: (a) mean absolute magnetization and (b) per-spin heat-capacity fluctuation estimate, Eq. (54). Dashed and dotted curves show the exact target and uniform encoder reference. Conditional IQP curves and bands show mean and sample standard deviation across five frozen confirmation circuits at 101 temperatures. Independent markers use separately fitted circuits at $\beta = 0.4, 1.0, 1.6$, with the same summaries. We compute nonlinear observables within each run before aggregation. The sweep includes fitting, development and extrapolation conditions. For learned distributions, panel (b) measures energy fluctuations rather than a verified temperature response.

dent baseline. Thus, capturing the change in magnetization magnitude does not ensure recovery of the target's energy fluctuations, and the relative accuracy depends on the observable and temperature.

D. Conditional signed-graph generation with exact switching symmetry

We use signed graphs to demonstrate conditional generation with an exact symmetry. Each edge carries a positive or negative sign. A vertex switch reverses the signs of all edges incident on that vertex. This operation preserves every triangle product because it reverses either zero or two signs within each triangle. We construct a temperature-dependent input that preserves this symmetry throughout training and conditional generation.

Sixteen qubits represent the edge signs of a seven-vertex graph. We connect every pair among vertices $0, \ldots, 4$, and connect each of vertices 5 and 6 to vertices $0, 1, 2$. Each edge bit x_{ij} specifies a sign $s_{ij} = (-1)^{x_{ij}}$. At inverse temperature β, we use the Gibbs target

$$p_\beta(x) = Z_\beta^{-1} e^{\beta E(x)}, \qquad E(x) = \sum_{\{i,j,k\} \in \mathcal{T}} s_{ij} s_{ik} s_{jk}, \tag{55}$$

where $\mathcal{T}$ contains the graph's sixteen triangles and Z_β normalizes the distribution. We call a triangle balanced when the product of its three edge signs is positive. Increasing β therefore favors graphs with more balanced triangles.

We describe combinations of vertex switches using signs $h_i \in \{-1, +1\}$, where $h_i = -1$ indicates a switch at vertex i. Their combined action replaces each edge sign s_{ij} by $h_i s_{ij} h_j$. Switching all seven vertices reverses every edge twice and leaves the graph unchanged. We therefore fix $h_0 = 1$ to remove duplicate representations. The remaining six binary choices give $2^6 = 64$ distinct switching transformations, including the identity. We denote their corresponding edge-bit masks by G; applying a mask $g \in G$ flips the indicated bits, $x \mapsto x \oplus g$. Every transformation preserves the target distribution at every temperature.

We encode temperature using ten triangles that share vertex 0. The six root edges $0i$, with $i = 1, \ldots, 6$, form a spanning tree: they connect all seven vertices without forming a cycle. We call the remaining ten edges *chords* and denote their set by $\mathcal{C}$. Each chord ij, together with root edges $0i$ and $0j$, forms a triangle $(0, i, j)$.

We initialize all sixteen qubits in $|0\rangle$. For each chord ij, we apply $R_X(t_\beta)$ to its qubit, then use that qubit to control two CNOT gates targeting root edges $0i$ and $0j$. On an isolated triangle initially in $|000\rangle$, this motif creates a superposition of $|000\rangle$ and $|111\rangle$ (Figure 6). In either component, each vertex touches zero or two edges with bit value one, giving even incidence parity.

In the full graph, the triangles share root-edge qubits. We repeat the rotation-and-CNOT pattern for all ten chords, combining the shared root-edge bits modulo two. This preserves even incidence parity at every vertex. Finally, we apply a single Hadamard layer to all sixteen qubits.

After the Hadamard layer, the encoded input takes the form

$$|\phi_\beta\rangle = \exp\left[-\frac{i t_\beta}{2} \sum_{(i,j) \in \mathcal{C}} Z_{0i} Z_{0j} Z_{ij} \right] |+\rangle^{\otimes 16}, \tag{56}$$

$$t_\beta = \frac{\pi}{2} \tanh(1.5\beta). \tag{57}$$

We fix this temperature map throughout training. Each Pauli-Z product contributes a phase determined by the edge-sign product of its triangle. Because these operators are diagonal in the computational basis, they leave the uniform measurement probabilities of $|+\rangle^{\otimes 16}$ unchanged. Temperature therefore enters entirely through correlated phases, which the learned circuit must convert into changes in the output distribution.

We prepare the encoded input using ten rotations, twenty CNOTs and sixteen Hadamards, without ancillary qubits. For classical training, we use the state before the Hadamard layer as the reference. To sample its measurement distribution, we draw the ten chord bits independently, assigning each a value of one with probability $\sin^2(t_\beta/2)$. We then set each root-edge bit $0i$ to the sum, modulo two, of the chord bits incident on vertex i. The chord bits also determine each supported amplitude through a product of ten rotation coefficients, allowing us to evaluate the required amplitude ratios directly. Appendix C gives the explicit expressions.

The Hadamard layer converts the reference state's even incidence parity into invariance under vertex switches:

$$X^g \left|\phi_\beta\right\rangle = \left|\phi_\beta\right\rangle, \qquad g \in G, \tag{58}$$

where X^g flips the edge bits specified by the switching mask g. Every IQP layer generated by Pauli-X strings commutes with these switching operators and therefore preserves this invariance, regardless of its learned parameters.

For this experiment, we use two trainable angles:

$$U_{\alpha,\gamma} = \exp\left(i\alpha \sum_{i=1}^{6} X_{0i} + i\gamma \sum_{(i,j)\in\mathcal{C}} X_{ij}\right). \tag{59}$$

The angle α controls the six root-edge rotations, while γ controls the ten chord rotations. We share both angles across all temperatures. This layer adds sixteen local rotations and no entangling gates. The resulting model satisfies

$$q_{\alpha,\gamma}(x \mid \beta) = q_{\alpha,\gamma}(x \oplus g \mid \beta) \tag{60}$$

for every $g \in G$, throughout training and at unseen temperatures, without averaging the output over switches.

We train at $\beta \in \{0, 0.4, 0.8, 1.2\}$, using 128 target samples per temperature and 512 in total per fit. We optimize sampled MMD and use exact target distributions at separate development temperatures for model selection. After fixing the hyperparameters, we train five independent models.

We evaluate full-distribution TVD exactly by grouping configurations related by vertex switches, as detailed in Appendix C. Across twelve evaluation temperatures excluded from training and model selection, we obtain a mean TVD of 0.1780 ± 0.0013. We first average TVD over temperatures within each fit, then report the mean and sample standard deviation across the five fits. Table IV separates interpolation from extrapolation and reports the worst-temperature errors.

We characterize graph balance using two observables. Mean triangle balance, $b = \langle E\rangle/16$, averages the triangle-sign products over triangles and generated graphs. The probability of complete balance, $P_{\mathrm{bal}} = \Pr(E = 16)$, measures how often all sixteen triangle products are positive. Figure 7 compares the learned and target responses over $\beta \in [0, 2.4]$, together with the uniform encoder-only baseline. We evaluate this temperature sweep after model selection; it includes fitting and development temperatures.

The learned distributions increasingly favor balanced triangles as inverse temperature increases, but their response saturates below the target. At $\beta = 2.4$, the models achieve $b = 0.9148\pm0.0017$ and $P_{\mathrm{bal}} = 0.7983\pm0.0046$, compared with target values of 0.9999 and 0.9996, respectively. Across ten additional evaluation temperatures from 1.5 to 2.4, beyond the original evaluation range, we obtain a mean full-distribution TVD of 0.1981 ± 0.0046.

To assess the role of conditioning, we hold each trained model's input temperature fixed at $\beta = 0.6$. This increases mean TVD across the twelve evaluation temperatures from 0.1780 ± 0.0013 to 0.4886 ± 0.0011, while the encoder-only uniform distribution gives 0.7275. The increase shows that the learned response depends on the encoded temperature. This control measures sensitivity to conditioning; it does not evaluate an independently optimized model with a fixed input.

The experiment demonstrates conditional generation with a classically accessible correlated reference and exact switching symmetry, using two learned angles. The current models still misrepresent the target's energy fluctuations at large inverse temperatures, and a fixed classical model that enforces switching symmetry achieves lower mean TVD. Appendix C presents these diagnostics. The results therefore establish the construction's symmetry capability without demonstrating a performance advantage over the classical control.

TABLE IV. Full-distribution TVD for the sixteen-qubit signed-graph experiment. Entries show mean ± sample standard deviation across five fits, each using 128 target samples per training temperature. We first average TVD over temperatures within each fit; the worst-temperature row instead uses each fit's maximum TVD. Parentheses give the number of temperatures. The twelve evaluation temperatures exclude training and model selection. Extended extrapolation covers ten additional temperatures from $\beta = 1.5$ to 2.4, evaluated after model selection.

Evaluation summary	TVD ↓
All evaluation temperatures (12)	0.1780 ± 0.0013
Interpolation (10)	0.1775 ± 0.0006
Extrapolation (2)	0.1806 ± 0.0046
Worst temperature	0.2981 ± 0.0018
Extended extrapolation (10)	0.1981 ± 0.0046

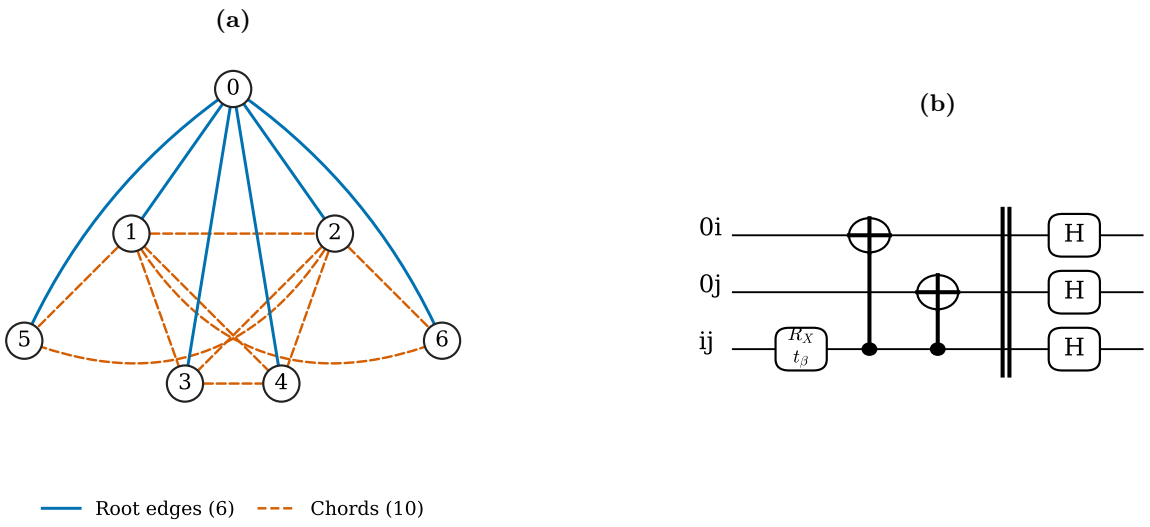


FIG. 6. Triangle-based temperature encoding. Panel (a) shows the seven-vertex graph, with six root edges forming a spanning tree and ten chords. Panel (b) shows the preparation motif for a fundamental triangle. A chord rotation $R_X(t_\beta)$ and two CNOTs prepare $\cos(t_\beta/2)\,|000\rangle - i\sin(t_\beta/2)\,|111\rangle$ on edges $0i, 0j, ij$. The full encoder repeats the rotation and CNOT construction for ten chords with shared root edges, then applies the global Hadamard layer once. The displayed motif is therefore an illustration, rather than an independent three-qubit block repeated with its own Hadamards. Every rotation uses $t_\beta = (\pi/2)\tanh(1.5\beta)$.

E. Scaling

We assess the computational feasibility of classical training at $n \in \{64, 128, 256, 512, 1024, 2048\}$ visible qubits, without ancillas. We use the fixed product-state encoder $R_Z(0.75\beta)R_Y(\pi/2)$ on every qubit. The learned unitary uses independent one-body phase gates and two-body phase gates between horizontal and vertical nearest neighbors on an open rectangular lattice. For an $L_x \times L_y$ lattice with $n = L_xL_y$, the model has $3n - L_x - L_y$ trainable parameters, so the parameter count grows linearly with n. Appendix A 4 gives the lattice dimensions and optimization settings.

At each system size, we train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$ and track the mean sampled MMD^2 over the held-out temperatures throughout optimization (Figure 8). Appendix A 4 specifies the training and reporting protocol.

The scaling study demonstrates the computational feasibility of classical conditional training up to 2048 qubits (Figure 8). At 64 qubits, the median held-out sampled MMD^2 decreases from 0.0504 to 0.0350 after 5,000 updates; at 2048 qubits, it decreases from 0.0490 to 0.0417. These modest reductions suggest limited progress under the chosen training configuration and may reflect suboptimal hyperparameters. The runs use a sparse lattice ansatz and optimizer settings that differ from the tuned sixteen-qubit comparison. Further tuning and benchmarking are needed for a systematic assessment of generative performance. Because the kernel bandwidths vary with system size, the absolute MMD^2 values do not directly compare distributional fidelity across sizes.

V. INPUT-STATE INDUCED SYMMETRIES IN IQP BORN MACHINES

Input-state encoding can also enforce exact output symmetries without restricting the learned phases. For a nonzero bit mask $s \in \{0,1\}^n$, the bit-flip operator $X_s = \bigotimes_{i=1}^n X_i^{s_i}$ commutes with the IQP unitary. If each encoded input is an eigenstate of X_s, the output state retains the same eigenvalue. Its computational-basis amplitudes at x and $x \oplus s$ therefore differ at most by a sign, giving equal probabilities for every condition and parameter vector. This applies the commuting-symmetry construction of Recio-Armengol *et al.* [28, Sec. 6] to conditional inputs. Such constraints provide inductive biases in variational quantum models [43].

The conditional switching experiment in Section IV D implements this criterion for six independent masks simultaneously. Its reference state has even incidence parity at every graph vertex. After the Hadamards, the input is a simultaneous $+1$ eigenstate of all vertex switches, so any subsequent X-generated IQP layer preserves the constraints. Parameter tying reduces the fitted model to two angles but is not required for the symmetry guarantee. This separates an architectural property from distributional accuracy, which still depends on the target, encoder and optimization.

VI. DISCUSSION

Our framework establishes sufficient conditions for classical training of conditional IQP Born machines with

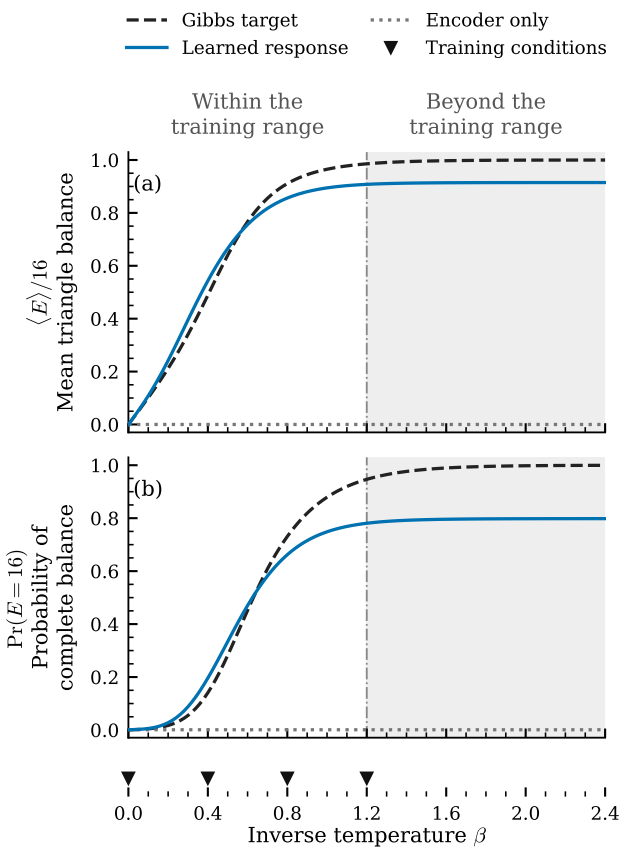


FIG. 7. Temperature dependence of signed-graph balance. Panel (a) shows the mean triangle balance $\langle E \rangle/16$, and panel (b) shows the probability that all sixteen triangle products are positive. Solid blue curves show means across five independently trained models, with bands indicating one sample standard deviation. Dashed black curves show the exact Gibbs target, and dotted gray curves show the encoder-only distribution, which remains uniform at every temperature. Downward triangles mark fitting temperatures; shading identifies temperatures beyond the fitting range. We evaluate the models after selection on 121 temperatures over $\beta \in [0, 2.4]$, including fitting and development conditions.

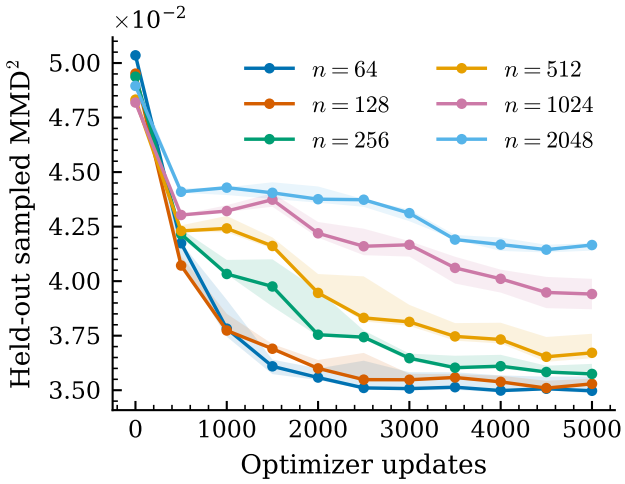


FIG. 8. Classical training of conditional IQP Born machines from 64 to 2048 qubits. At each plotted update, we average sampled MMD^2 over the held-out inverse temperatures $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$. Curves and shaded bands show the median and interquartile range of these averages across three independent training runs. We fix the expected mask weights at $w \in \{2, 4\}$ for each system size, so the corresponding kernel bandwidths vary with n.

input preparation beyond strict IQP form. Efficient sampling and amplitude-ratio evaluation of the reference state enable unbiased estimation of conditional training moments and their gradients, while one learned unitary shares parameters across conditions. The sixteen-qubit experiments demonstrate conditional translations and changes in distributional shape through condition-dependent input phases. The two-dimensional Gaussian experiment approximates mean–width combinations absent from fitting and development, including pairs outside the joint training hull of separately varied coordinates. Its phase law supplies a structured response prior that one shared angle calibrates. Finite-target validation preserves useful mean and width responses, while direct calibration from fitting-pool means achieves comparable or lower error without iterative optimization. A product-prior control also matches or improves fidelity. We therefore interpret this benchmark as structured conditional generation and calibration, without claiming that training discovers the response law or that Gaussian amplitude preparation is necessary. For Curie–Weiss, the shared model achieves lower mean full-distribution TVD than both independent baseline families at $\beta = 1.0$ and 1.6, with the latter extending beyond the training range. Each baseline family fits these temperatures directly and uses three times the aggregate parameters and optimizer updates, whereas the shared model uses more training-pool samples. At $\beta = 0.4$, the uniform encoder-only distribution achieves the lowest error among the models in the primary comparison. Joint training may provide useful regularization, but comparisons with matched data and optimization budgets are needed to assess this explanation.

Input-state design can impose exact output symmetries while preserving classical trainability. The conditional signed-graph construction guarantees sixty-four switching translations for every learned parameter vector. Five independent finite-data fits give mean reporting TVD 0.1780 ± 0.0013 with two learned angles, while the graph-order response extends beyond the fitting range. This symmetry guarantee does not ensure a faithful Gibbs response: the learned balance saturates below the target and high-inverse-temperature energy fluctuations remain inaccurate. A fixed classical switching control achieves lower reporting TVD, and the model admits exact quotient evaluation. These results support corre-

lated conditional encoding and symmetry control without establishing a classical fidelity or computational advantage.

The methodological limitations concern distributional evaluation and the scope of the scaling evidence. Efficient, unbiased MMD estimation does not ensure accurate recovery of every relevant correlation [44]. Although the Gaussian kernel includes all correlation orders, its bandwidth weights them unequally, and finite sampling and optimization may fail to resolve weakly weighted discrepancies. Agreement in Hamming-weight profiles can therefore coexist with errors in individual bitstring probabilities, making full-distribution diagnostics necessary whenever feasible. The observable sweep further shows that recovering magnetization trends does not ensure accurate energy fluctuations, especially beyond the fitting range. For Curie–Weiss, reporting reuses the confirmation runs used to select hyperparameters, so the reported spread does not independently assess model selection. The reporting temperatures are excluded from fitting and selection but were examined in earlier experiments. Independent final fits and previously unexamined conditions, evaluated with configurations fixed in advance, would strengthen the generalization evidence. The sparse-lattice runs establish computational feasibility up to 2048 qubits, with modest reductions in held-out sampled MMD^2. Their connectivity and optimizer settings differ from the sixteen-qubit comparison. Fixed expected mask weights emphasize low-order correlations as the system grows, while size-dependent bandwidths prevent direct comparisons of distributional accuracy across sizes.

Future work should examine richer tractable encoders and optimization strategies that improve conditional accuracy. Broader families of correlated inputs and multidimensional encodings should be assessed alongside state-preparation and estimation costs, with particular attention to calibration of each conditional response. Conditional extensions of spectral Born machines would require corresponding sampling, amplitude-ratio and gradient estimators [45]. Generator selection, initialization and adaptive kernels merit investigation [39–41], including whether gradients from different conditions reinforce or cancel one another. Quantum deployment should assess how noise and circuit partitioning affect the conditional response and imposed symmetries [46, 47]. Practical quantum sampling advantage remains open: our Gaussian, Curie–Weiss and signed-graph targets admit efficient classical sampling. Demonstrations would require comparisons with strong classical generative models and sampling-hardness or verification arguments appropriate to the trained conditional circuits.

VII. CONCLUSIONS

We introduced a conditional framework for IQP Born machines that combines a shared learned unitary with encoders that need not have IQP form. Efficient reference-state sampling and amplitude-ratio evaluation enable unbiased classical estimation of conditional training observables and their gradients, allowing optimization without quantum circuit sampling. Input-state design also lets us impose exact output symmetries independently of the learned parameters.

Our sixteen-qubit experiments demonstrate conditional translations and changes in distributional shape through input phases. The two-dimensional Gaussian model generalizes to joint mean–width combinations absent from fitting and development using one angle shared across sixteen local rotations. For the Curie–Weiss model, the shared model achieves lower mean full-distribution TVD than both independently fitted baseline families at two held-out temperatures, including one beyond the training range, with fewer aggregate parameters and optimizer updates. The signed-graph experiment demonstrates finite-data conditional generation with exact switching symmetry and two learned angles. Additional runs establish the computational feasibility of classical conditional training up to 2048 qubits. Together, the theoretical constructions and numerical results establish conditional generation and symmetry control within a framework that supports classical training. Further work must assess distributional accuracy at larger scales and practical quantum advantage.

DATA AVAILABILITY

The code and data supporting this study are available on GitHub [48].

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Appendix A: Hyperparameters and evaluation protocols

1. Common training settings

We train all models using the unbiased estimator of the squared-MMD objective. We keep the encoders fixed and initialize the circuit phases independently as $\theta_j \sim \mathcal{N}(0, s_\theta^2)$. At each update, we sample one fitting condition and one kernel bandwidth uniformly, then draw a target minibatch with replacement. We draw fresh nonidentity masks and apply the probability correction for excluding the identity mask. Mask budgets count draws, including repeated masks. We draw reference samples independently of the target minibatch and reuse them across masks. Table V summarizes the translation, Curie–Weiss and scaling settings. Appendices B and C give the separate two-dimensional Gaussian and conditional switching protocols.

For each initialization, we reset the optimizer and draw new circuit phases while reusing the target pools. For the experiments summarized in Table V, each training run includes all prescribed initializations, and we select the checkpoint with the lowest selection score across them. Reported uncertainty summarizes variation across independent training runs.

We train the Gaussian and scaling models with `Adam` [49], using cosine learning-rate decay to one tenth of the initial rate. For the Curie–Weiss models, we use the optimizer selected by HPO and keep the learning rate constant. We set $b_1 = 0.9$, and use $b_2 = 0.999$ and $\epsilon = 10^{-8}$ for the `Adam` variants. We set weight decay to zero.

TABLE V. Training settings and budgets per circuit. Mask and reference budgets give the number of draws per update. We select Curie–Weiss hyperparameters and checkpoints using full-distribution TVD. Table VII gives the search space for the optimizer, learning rate and initialization width. We report selected learning rates and initialization widths to three significant figures and display the expected mask weight 7.99 as 8. For Curie–Weiss, the update and initialization budgets apply to confirmation and evaluation; screening uses the smaller budgets specified in Sec. A 3 a.

Setting	Gaussian translation	Shared product input	Independent conditional	Independent unconditional	Scaling
Updates per initialization	1,024	2,000	2,000	2,000	5,000
Initializations per run	1	3	3	3	3
Reported runs	5	5	5	5	3
Training targets per condition	512	1,024	1,024	1,024	10,000
Target minibatch	128	256	256	256	450
Training mask draws	1,024	2,048	2,048	2,048	512
Training reference draws	1,024	2,048	2,048	2,048	1,024
Expected mask weights	$\{1\}$	$\{2, 4, 8\}$	$\{2, 4, 8\}$	$\{2, 4, 8\}$	$\{2, 4\}$
Checkpoint interval	32	100	100	100	25
Optimizer	`Adam`	`Adam`	`AdamW` `ScheduleFree`	`Adam`	`Adam`
Learning rate	0.005	0.0310	0.0603	0.00844	0.01
Learning-rate schedule	Cosine	Constant	Constant	Constant	Cosine
Initialization width s_θ	0.25	0.696	0.119	0.117	0.1
Coefficient b_1	0.9	0.9	0.9	0.9	0.9

For an expected mask weight w, we set the Gaussian bandwidth to

$$\sigma(n, w) = \left[-2 \ln\left(1 - \frac{2w}{n}\right)\right]^{-1/2}, \qquad 0 < w < n/2, \tag{A1}$$

where n denotes the number of visible qubits and w is the expected mask weight before excluding the identity mask. We weight the bandwidths equally in the objective. For the sixteen-qubit Curie–Weiss models, we use $w \in \{2, 4, 7.99\}$ and display 7.99 as 8. Using 7.99 keeps $w < n/2$ and the bandwidth strictly positive.

We report rotation angles in radians and compute sixteen-qubit full and profile TVDs from exact circuit probabilities. Sampled-MMD diagnostics do not enter these calculations.

2. Discrete Gaussian translation

We use 16 visible qubits on an open 4×4 lattice. The learned unitary contains 16 one-body phase gates and 24 two-body phase gates between horizontal and vertical nearest neighbors. We prepare the input state with the fixed product-state encoder

$$|\phi_{\mathrm{G}}(c)\rangle = \bigotimes_{i=1}^{16} R_Z(c-1/2)R_Y(1.5)\,|0\rangle_i\,. \tag{A2}$$

We choose the near-equatorial angle 1.5 radians empirically to avoid an input state whose output distribution the learned unitary cannot change. At $c = 0.5$, replacing this angle with $\pi/2$ would prepare $|+\rangle^{\otimes 16}$, a simultaneous eigenstate of every X-type generator in the IQP unitary. The learned parameters would then contribute only a global phase, leaving a uniform output distribution with a fixed binomial Hamming-weight profile. Using 1.5 removes this exact invariance and allows the learned parameters to modify the output distribution at $c = 0.5$.

We use the target Hamming-weight profile $\rho(k \mid c) \propto \exp[-(k-(5+6c))^2/8]$, where $k = |x|$. Within each Hamming-weight shell, we assign equal probability to all bitstrings. We train at $c \in \{0, 0.25, 0.5, 0.75, 1\}$, using 512 target samples per condition and 2560 per run.

We evaluate the model at initialization and every 32 updates through update 1024, selecting the checkpoint with the lowest mean exact profile TVD over the development conditions $c \in \{0.125, 0.375, 0.625, 0.875\}$. We then evaluate the selected model at 80 held-out conditions, $c_j = (j + 1/3)/80$ for $j = 0, \ldots, 79$. We draw 1000 target samples per evaluation condition for auxiliary sampled-MMD diagnostics.

We compute full-distribution and profile TVDs from exact circuit probabilities. For each run, we average each error over the evaluation conditions and report the median of these averages across five independent runs. Figure 2 averages exact profiles across the five selected circuits at the development conditions and shows the median and interquartile range of their mean output Hamming weights at each held-out condition.

3. Curie–Weiss models

We use the sixteen-qubit target $p(x \mid \beta) \propto \exp[\beta(2|x| - 16)^2/32]$. For all trained sixteen-qubit Curie–Weiss models, we parameterize the learned unitary with 16 one-body phase gates and 120 two-body phase gates, without ancillas. For the shared product-input model, we prepare $R_Z(0.75\beta)R_Y(\pi/2)\,|0\rangle$ on every qubit.

We train the shared product-input model at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$, use $\beta \in \{0.5, 0.7, 1.3, 1.5\}$ for model selection, and report performance at $\beta \in \{0.4, 1.0, 1.6\}$. Each independent baseline family comprises three circuits that we fit directly at these reporting temperatures, using either the corresponding product input or $|0\rangle^{\otimes 16}$. For each baseline family, we select one optimizer, learning rate and initialization width jointly for its three circuits.

We train each circuit using 1024 target samples per fitting temperature and run, minibatches of 256, and 2048 mask draws and 2048 reference draws per update. Table VI summarizes the aggregate fitting resources.

For auxiliary sampled-MMD diagnostics, we use 2048 validation targets per fitting temperature and 2048 independently drawn test targets per reporting temperature. Each diagnostic uses 2048 mask draws per condition and bandwidth, and 2048 reference samples per condition shared across bandwidths. We do not use these diagnostics for model selection.

TABLE VI. Curie–Weiss fitting budgets per run for confirmation and evaluation. Training-target counts sum the pool entries over fitting temperatures and exclude repeated minibatch draws. Optimizer-update counts include all three initializations of every circuit. The shared model fits one circuit, and each independent baseline family fits three.

Model	Trained circuits	Training targets	Optimizer updates
Shared product input	1	4,096	6,000
Independent conditional	3	3,072	18,000
Independent unconditional	3	3,072	18,000

a. *Hyperparameter optimization*

We report one Optuna study for each of the three Curie–Weiss model families, using a tree-structured Parzen estimator (TPE) [50]. Each study evaluates 60 configurations. We tune the optimizer choice among `Adam`,

`AdamWScheduleFree` and `SGDScheduleFree`, the constant learning rate, and the Gaussian initialization width [49, 51]. Table VII gives the search space. We assign each family the same number of trials, although each independent-family trial fits three circuits per run.

We use random search as a reference for the trial budget [52]. For 60 independent configurations drawn from a fixed sampling distribution, the probability of sampling at least one configuration in its best-performing 5% is $1 - 0.95^{60} \approx 0.954$. This calculation motivates the budget but does not establish equivalent coverage for adaptive TPE proposals.

TABLE VII. Curie–Weiss hyperparameter search space. We treat optimizer choice as categorical and search learning rates and Gaussian initialization widths on a logarithmic scale.

Hyperparameter	Options or range
Optimizer	`Adam`, `AdamWScheduleFree`, `SGDScheduleFree`
Learning rate, `Adam` variants	$[3 \times 10^{-4}, 10^{-1}]$
Learning rate, `SGDScheduleFree`	$[10^{-3}, 1]$
Initialization width s_θ	$[0.01, 0.7]$ radians

We begin each study with 12 startup trials, including one reference configuration per optimizer. We set the reference learning rates to 0.01 for the `Adam` variants and 0.1 for `SGDScheduleFree`, with $s_\theta = 0.5$ for all reference configurations. We do not prune trials.

We fix the optimizer coefficients as specified in the common training settings and use no warmup or gradient clipping. For schedule-free optimizers, we set the learning-rate weighting exponent to 2 and score checkpoints using the averaged parameters provided for evaluation.

For each screening trial, we train every circuit for 1000 updates on each of three runs, using one initialization per run. We compute exact TVD from the full distribution over the 2^{16} computational-basis states at initialization and every 100 updates. For each shared model, we select the checkpoint with the lowest mean TVD over the development temperatures. Within each independent baseline family, we select each circuit's checkpoint using TVD at its fitted temperature and average the three selected errors within each run. We then take the median of the three run scores as the trial score.

We confirm the three best distinct configurations from each study on five fresh runs, training each circuit for 2000 updates from each of three initializations per run. We select each circuit's checkpoint across its initializations using the same TVD criterion as in screening, then choose the configuration with the lowest median score across the five runs.

We report performance using the five selected confirmation runs, without further fitting or checkpoint selection. Because these runs also inform hyperparameter selection, they do not provide an independent final evaluation. In the current protocol, we exclude the shared models' reporting temperatures from fitting and selection, although earlier experiments examined them. We fit the independent circuits directly at their reporting temperatures. For each temperature, we report the mean and sample standard deviation across the five runs.

TABLE VIII. Exact Hamming-weight profile TVD for sixteen-qubit Curie–Weiss models. We report means ± sample standard deviations across the five confirmation runs used for model selection and reporting. We select models using full-distribution TVD; encoder-only values are deterministic. We evaluate the shared model at temperatures excluded from fitting and selection, and fit each independent circuit at its reporting temperature. Bold entries identify the lowest mean error at each temperature, including the encoder-only baseline.

Model	Evaluation	Hamming-weight profile TVD ↓		
		$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Encoder only	Baseline	0.1180	0.4406	0.8699
Shared conditional IQP	Held-out	0.0976 ± 0.0269	0.1301 ± 0.0317	$\mathbf{0.0631 \pm 0.0135}$
Independent conditional IQPs	Fitted	$\mathbf{0.0939 \pm 0.0073}$	0.0885 ± 0.0116	0.3948 ± 0.0003
Independent unconditional IQPs	Fitted	0.1186 ± 0.0040	$\mathbf{0.0780 \pm 0.0320}$	0.3293 ± 0.0119

Table VIII reports exact Hamming-weight profile errors for the primary Curie–Weiss comparison.

b. Physical observables

For a distribution q, let $\rho_q(k \mid \beta)$ denote its exact Hamming-weight profile and set $m_k = (2k-n)/n$ with $n = 16$. We compute its moments as $\langle m^r \rangle_q = \sum_{k=0}^{n} \rho_q(k \mid \beta) m_k^r$, and its mean absolute magnetization as $\sum_k \rho_q(k \mid \beta)|m_k|$. The target law corresponds to $H = -nm^2/2$ in units $J = k_B = 1$. Excluding self-interactions would add only a constant to H and would not change the energy variance. Equation (54) therefore becomes

$$c_{\mathrm{fluc},q}(\beta) = \frac{\beta^2 n}{4} \left(\langle m^4 \rangle_q - \langle m^2 \rangle_q^2 \right). \tag{A3}$$

The observables depend only on the shell profile and do not test uniformity within shells. For the Gibbs target, differentiating the mean energy per spin with respect to $T = \beta^{-1}$ gives this fluctuation expression. We do not assume that the learned family satisfies the same derivative identity.

We evaluate the five selected shared product-input circuits on 101 equally spaced inverse temperatures $\beta_j = 0.02j$, $j = 0, \ldots, 100$, with parameters fixed after selection. The grid includes fitting and development temperatures, as well as extrapolation below and above the fitting range. These are the same confirmation runs used for hyperparameter selection, so the sweep is not an independent assessment of model selection. We compute each observable within each run before taking its mean and sample standard deviation, including the nonlinear heat-capacity estimate. The independent baselines appear only at their fitted temperatures, with the same calculation and aggregation. The uniform encoder reference has the binomial shell law $\rho_0(k) = 2^{-n}\binom{n}{k}$.

4. Scaling

We use the product-state Curie–Weiss encoder on every qubit, with no ancillas, and adopt the open lattice architectures listed in Table IX. At size n, the target distribution is $p(x \mid \beta) \propto \exp[\beta(2|x| - n)^2/(2n)]$.

We train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$, using 10,000 target samples per condition and 40,000 per run. We evaluate at $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$ and use three independent runs at each size. We use the cosine-decay `Adam` settings in Table V, with 5000 updates per initialization and three initializations per run. These settings differ from the selected sixteen-qubit configurations. We have not tested the transfer of those configurations or retuned the scaling models.

TABLE IX. Open rectangular lattice architectures for the scaling experiments. We assign one trainable phase to each qubit and one to each horizontal or vertical nearest-neighbor pair.

Visible qubits	Lattice	Trainable parameters
64	8×8	176
128	8×16	360
256	16×16	736
512	16×32	1,488
1,024	32×32	3,008
2,048	32×64	6,048

We select the checkpoint with the lowest mean sampled validation MMD^2 across training temperatures and bandwidths. We evaluate initialization and every 25 updates across three initializations. We exclude held-out temperatures from fitting and selection.

For each run, we report the selected initialization at update 0 and every 500 updates through update 5000. At each reporting update, we average five independent MMD^2 estimates over the evaluation temperatures and bandwidths. Each estimate uses 2000 target draws per condition, 512 mask draws per condition and bandwidth, and 1024 reference draws per condition. We reuse reporting sample sets across updates and share reference samples across bandwidths. Figure 8 shows the median and interquartile range across the three runs. We retain negative unbiased MMD^2 estimates.

Appendix B: Two-dimensional Gaussian encoder and evaluation protocol

1. Fitting, development and reporting conditions

For joint variation, fitting uses $\{-1, 0, 1\}^2$ and development uses

$$\{-1, -0.5, 0, 0.5, 1\} \times \{-0.75, -0.25, 0.25, 0.75\}.$$

The reporting grid is

$$u \in \{-1, -0.83, -0.37, 0.19, 0.61, 1\},$$
$$v \in \{-1, -0.79, -0.31, 0.23, 0.67, 1\}.$$

We exclude its four fitted corners from reporting averages, retaining thirty-two pairs.

Separate variation fits $(u, 0)$ at $u \in \{-1, -0.37, 0, 0.37, 1\}$ and $(0, v)$ at $v \in \{-1, -0.31, 0.31, 1\}$. Development uses $(u, 0)$ at $u \in \{-0.73, -0.17, 0.17, 0.73\}$ and $(0, v)$ at $v \in \{-0.69, -0.13, 0.13, 0.69\}$. Reporting uses sixteen mixed pairs,

$$\{-1, -0.37, 0.37, 1\} \times \{-1, -0.31, 0.31, 1\}.$$

Four satisfy $|u| + |v| \leq 1$ and lie inside the fitting hull; twelve lie outside. Every individual coordinate value appears in fitting, but no mixed reporting pair appears in fitting or development. We reuse previously explored condition sets, so this evaluates generalization of the fitted models rather than an independent confirmatory test of the selected encoder.

2. Sampled training and exact development selection

We fix the encoder before sampled training. A preceding exact-capacity pilot compared quadratic-phase coefficients $s \in \{0.2, 0.4, 0.8\}$ in $e^{-i(sv/16)J_z^2}$. It optimized the shared angle on fitting conditions and selected $s = 0.4$ using development TVD for both designs. This pilot also informed the nonzero initialization branch. The present experiment tests sampled calibration of this constructed family.

Each fit has a fixed pool of 1024 target bitstrings at each of nine fitting conditions, giving 9216 strings. Each update samples one condition and one bandwidth uniformly, draws 256 target strings with replacement, and draws 1024 fresh reference samples and 1024 nonidentity operator masks. We use expected mask weights $\{2, 4\}$ before conditioning on a nonidentity mask, with bandwidth $\sigma = (-2\ln(1 - 2w/16))^{-1/2}$ for expected weight w and the identity-mask probability correction. Conditional on the fixed pool, sampled MMD targets the empirical distribution. Adam uses $b_1 = 0.9$, $b_2 = 0.999$, $\epsilon = 10^{-8}$ and zero weight decay; cosine decay reduces the learning rate to one tenth of its initial value. We accumulate estimators in double precision, use single-precision matrix products and process reference draws in chunks of 256.

For each design, we screen four settings using 512 updates. Initialization is $\theta_0 \sim \mathcal{N}(m, 0.1^2)$ with $m \in \{0, \pi/3\}$, and the initial learning rate is 0.01 or 0.03. We retain the two settings with lowest exact development TVD, run two 2000-update confirmations per setting, and select the lower mean confirmation score. Both designs select $m = \pi/3$ and learning rate 0.03. Five final fits per design use seeds independent of configuration selection. The campaign comprises 26 fits, including screening and confirmation.

Checkpoint selection minimizes mean exact full-distribution TVD on the declared development conditions. We evaluate initialization, every 64 updates during screening or 100 updates otherwise, and the final update. Thus configuration and checkpoint selection have access to exact development target distributions in addition to the sampled fitting pools. The reported errors do not represent a pipeline restricted to those pools alone.

We pair pools and random streams across settings and designs; the stored numerical records retain their identifiers. One separate-variation fit returns its initialization. Initializations near zero do not reliably recover the desired width direction, so these results do not establish robustness across initialization branches.

We verify samples, amplitude ratios, scalar losses and scalar gradients against the original encoder representation. At four qubits, 256 independent estimates agree with a dense empirical-pool MMD oracle within one standard error for both loss and gradient. These checks establish implementation agreement at the tested settings; they do not quantify production-scale rare-event tails. Frozen source, pool and checkpoint hashes support replay of all 26 fits. Selected sixteen-qubit probabilities agree with the public circuit oracle at three checked pairs per final fit.

3. Reference access and preparation resources

The encoder amplitude for a bitstring of weight k is

$$a_{u,v}(k) = \sqrt{\frac{r_0(k)}{\binom{16}{k}}} \exp\left[\frac{iu(k-8)}{2} - \frac{iv(k-8)^2}{40}\right]. \tag{B1}$$

For a reference bitstring of weight j, the Hadamard-transformed amplitude is

$$\psi_{u,v}(j) = 2^{-8} \sum_{k=0}^{16} a_{u,v}(k) \mathcal{K}_k(j), \tag{B2}$$

$$\mathcal{K}_k(j) = \sum_{\ell=0}^{k} (-1)^\ell \binom{j}{\ell} \binom{16-j}{k-\ell}, \tag{B3}$$

with invalid binomial coefficients set to zero. We tabulate seventeen amplitudes, draw j with probability $\binom{16}{j}|\psi_{u,v}(j)|^2$, then draw a uniform weight-j bitstring. For a sample z and mask b, we evaluate the amplitude ratio using the entries at $|z|$ and $|z \oplus b|$. We divide only on sampled support. For general n, the table, transform and sampling operations require polynomial work. Concentrated reference laws can still produce rare, large ratios; tractable access does not imply small estimator variance for every condition.

The fixed prior admits a polynomial counter-based preparation. Write the per-string probability as $g(k) = r_0(k)/\binom{n}{k}$, set $F(n,k) = g(k)$, and recursively compute $F(i,k) = F(i+1,k) + F(i+1,k+1)$. At prefix length i and weight k, a counter-controlled rotation of angle $2\arcsin\sqrt{F(i+1,k+1)/F(i,k)}$ prepares the next bit. We update the counter after each visible bit and uncompute it at the end. At sixteen qubits this uses five clean workspace qubits, 136 controlled rotations and 160 counter-update or uncomputation operations. These are high-level operation counts, not compiled gate depth. Small-register preparation checks support this construction; the sampled campaign uses classical reference access and explicit state-vector preparation for circuit validation.

The remaining conditional preparation applies sixteen $R_Z(u/2)$ gates and 120 pairwise ZZ rotations. Since

$$J_z^2 = \frac{n}{4} I + \frac{1}{2} \sum_{i<j} Z_i Z_j,$$

$e^{-i(v/40)J_z^2}$ uses IsingZZ$(v/40)$ on every pair, up to a global phase. The learned layer adds sixteen $R_X(-2\theta)$ rotations. Efficient reference access and physical preparation are distinct resource requirements; this experiment does not measure hardware depth or deployment fidelity.

4. Evaluation, conditional response and fixed controls

We compute exact output probabilities in the seventeen-dimensional Dicke basis. Permutation symmetry makes full TVD equal to shell-profile TVD. We average over reporting conditions within each fit, then calculate the mean and sample standard deviation across five fits. The two designs have different reporting sets, so their table columns do not define a paired accuracy ranking.

After selection we also evaluate all 1681 pairs on the grid $u, v \in \{-1, -0.95, \ldots, 0.95, 1\}$. Figure 3 includes fitting and development conditions and shows the pointwise arithmetic mean of exact TVDs, rather than the TVD of a mixture. We display directly evaluated cells without smoothing. This dense sweep and the response curves are post-selection diagnostics.

The mean endpoint response is 0.961 after separate variation and 0.973 after joint variation, relative to the exact target displacement between $u = -1$ and $u = 1$ at $v = 0$. Width endpoint ratios at $u = (-1, 0, 1)$ are $(0.948, 0.913, 0.948)$ and $(0.945, 0.909, 0.945)$, respectively. We average each fit's ratio of output to target standard-deviation change between $v = -1$ and $v = 1$. None of the ten fits reverses an adjacent width step on these sweeps. These endpoint diagnostics do not imply perfect calibration throughout the domain.

The structured input already provides much of the response. Let $J = 8$, $m = J - k$, $a_m = \sqrt{r_0(J-m)}$ and $b = v/40$. Direct evaluation gives

$$A(b) = \sum_{m=-J}^{J-1} a_{m+1} a_m \sqrt{(J-m)(J+m+1)} \times \cos[b(2m+1)], \tag{B4}$$

$$\mathbb{E}[K] = 8 + \sin(2\theta) A(b) \sin(u/2). \tag{B5}$$

Matching the local mean slope to the nominal target slope 3 gives the analytic angle

$$\theta_{\mathrm{a}} = \tfrac{1}{2}\{\pi - \arcsin[6/A(0)]\} \simeq 1.146708.$$

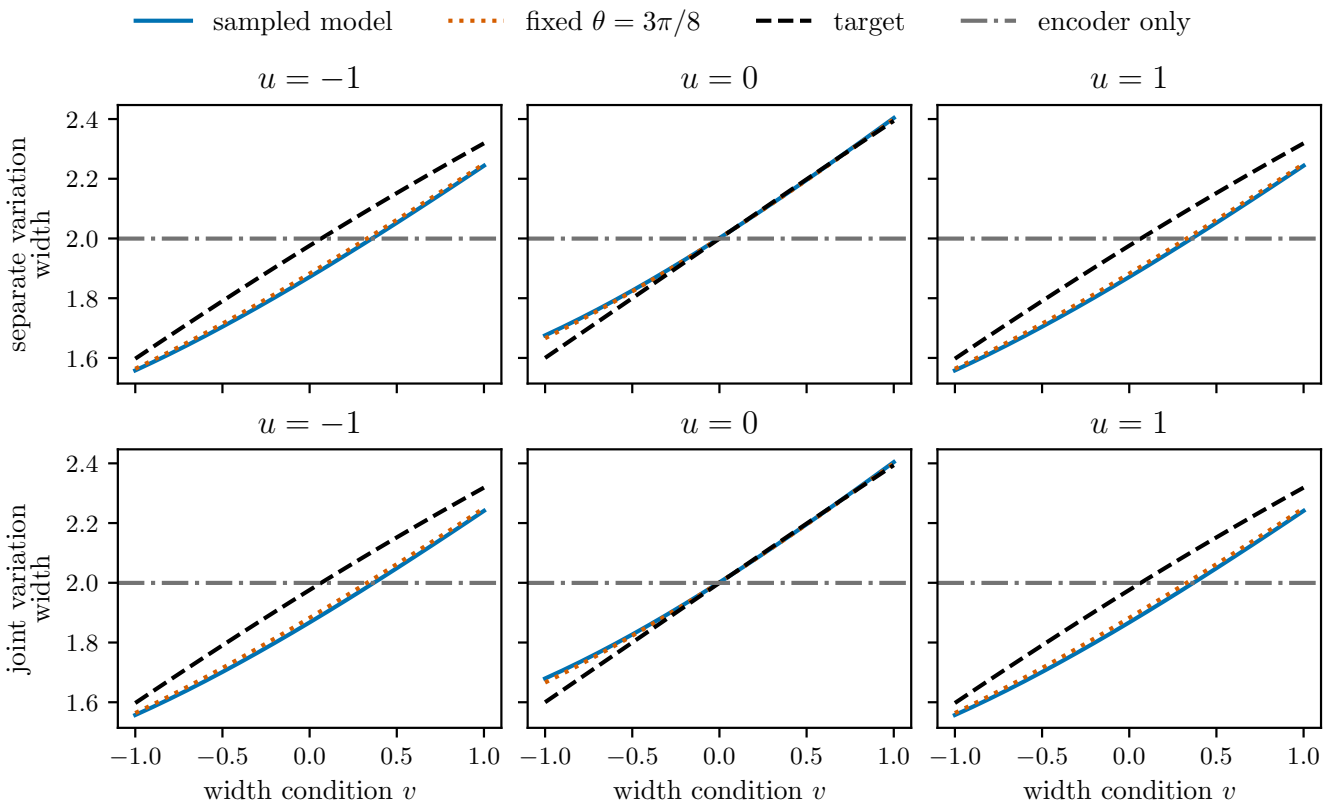


FIG. 9. Width response of the phase-conditioned Gaussian model. Columns fix $u = -1, 0, 1$; rows show separate and joint variation. Blue curves and bands show the pointwise median and interquartile range across five selected fits. Dashed black curves show exact target standard deviations, dotted orange curves use the fixed rotation $\theta = 3\pi/8$, and dash-dotted gray lines show the condition-independent encoder width. All displayed model slices increase with v. These 21-point sweeps include previously examined conditions and are diagnostic.

TABLE X. Selection and fixed-control diagnostics on the same reporting sets as Table II. Initial, selected and terminal rows summarize the same five sampled fits per design. The analytic controls use target-slope information, were derived after the campaign and underwent no training. They are descriptive diagnostics. The product prior retains the quadratic entangling phase.

	Separate variation 16 reporting pairs	Joint variation 32 reporting pairs
Initial checkpoint	0.0733 ± 0.0352	0.0682 ± 0.0335
Selected checkpoint	0.0258 ± 0.0005	0.0222 ± 0.0013
Terminal checkpoint	0.0364 ± 0.0037	0.0304 ± 0.0049
Fixed $\theta = 3\pi/8$	0.0347	0.0309
Analytic angle, Gaussian prior	0.0240	0.0215
Analytic angle, product prior	0.0217	0.0187

This controls the local slope rather than reproducing the entire discretized target law. Its reporting TVD matches or improves on the sampled selected means (Table X).

The fixed prior has squared overlap 0.999806 with $|+\rangle^{\otimes 16}$. Replacing it with that product state, while retaining both phases and the analytic angle, gives still lower diagnostic TVD. The quadratic phase continues to introduce entanglement for nonzero v; the replacement removes the Gaussian amplitude preparation, not the conditional pair gates. We derived these analytic controls after the campaign using target information and did not train the product replacement. The $3\pi/8$ control was frozen before sampled training after the capacity pilot. These comparisons support a constructed conditional-capability example, without establishing a benefit from the Gaussian prior or a necessity for stochastic learning.

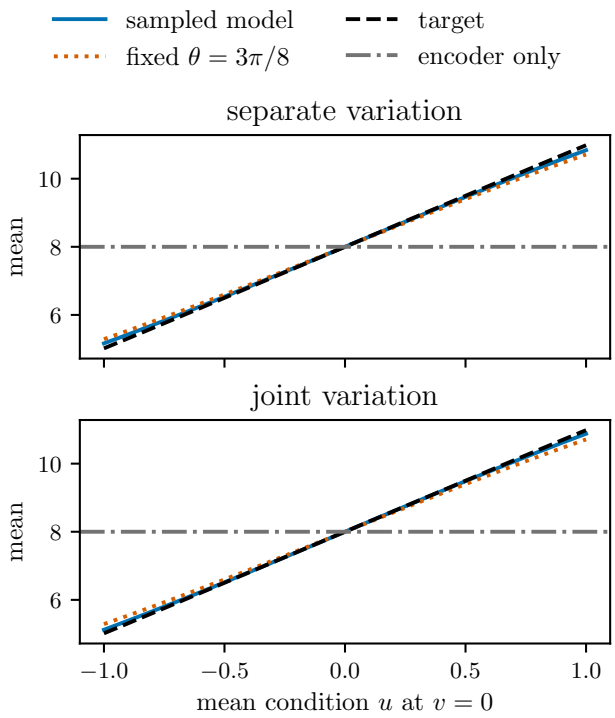


FIG. 10. Mean response at $v = 0$ for the same five selected fits per design. Curves, bands and controls follow Figure 9. The fixed encoder mean is eight for every condition.

Terminal-checkpoint reporting errors are 0.0304 ± 0.0049 for joint variation and 0.0364 ± 0.0037 for separate variation, compared with selected errors 0.0222 ± 0.0013 and 0.0258 ± 0.0005. The development oracle therefore contributes materially to the reported fidelity. We retain both outcomes when interpreting sampled calibration and do not attribute selected performance to the fitting samples alone.

5. Finite-target validation and direct calibration

We repeat sampled training with the previously selected encoder and optimizer, using five paired fits per design for the original estimator and a bounded pair-averaged estimator. The twenty fits share fitting and validation pools, initialization and random streams within each estimator pair. We keep the same fitting, development and reporting geometries. Each fit uses 9216 fixed fitting strings and 128 independent validation strings per development condition, totaling 1024 additional strings for separate variation or 2560 for joint variation. We evaluate reporting conditions only after all fits finish. The fresh pools do not remove the earlier target-informed configuration choices or previous examination of reporting conditions.

We select among initialization and twenty subsequent checkpoints using the model-dependent part of empirical validation MMD. Let $q_c(k)$ denote the model's shell profile, $\widehat{p}_c(k)$ the empirical validation profile and $\mathcal{K}_{kl}$ the kernel averaged over uniformly drawn bitstrings of weights k, l. The selection score averages

$$q_c^T \mathcal{K} q_c - 2 q_c^T \mathcal{K} \widehat{p}_c$$

over development conditions. Shell projection preserves the model-dependent terms because q_c is uniform within each shell. The empirical target self-term is independent of the learned angle and can be omitted when ranking checkpoints. We evaluate model moments exactly in the seventeen-dimensional symmetric subspace; target information comes only from the finite validation strings. Training remains sampled. Thus this protocol uses finite-target selection with exact model evaluation, rather than fully sampled validation.

The original estimator gives selected reporting TVD 0.0288 ± 0.0058 for separate variation and 0.0280 ± 0.0035 for joint variation. Terminal errors are 0.0332 ± 0.0053 and 0.0350 ± 0.0043. One separate-variation fit selects initialization. Mean per-fit worst-reporting-condition TVD is approximately 0.042 in both designs. Selected mean-displacement ratios are 1.005 and 1.007; width-change ratios at $u = (-1, 0, 1)$ are approximately $(0.933, 0.892, 0.933)$

TABLE XI. Full-distribution TVD with finite-target validation and exact model moments. Values are means and sample standard deviations across five fits on each design's reporting set. The designs have different reporting conditions. Direct calibration uses fitting-pool means only, without validation data or iterative optimization.

Model	Separate variation	Joint variation
Original estimator, selected	0.0288 ± 0.0058	0.0280 ± 0.0035
Original estimator, terminal	0.0332 ± 0.0053	0.0350 ± 0.0043
Pair-averaged selected	0.0289 ± 0.0057	0.0279 ± 0.0029
Pair-averaged terminal	0.0336 ± 0.0049	0.0349 ± 0.0043
Direct calibration, Gaussian prior	0.0263 ± 0.0025	0.0259 ± 0.0019
Direct calibration, product prior	0.0241 ± 0.0026	0.0230 ± 0.0019

in both designs. No selected fit reverses an adjacent width step on the checked slices. Clamping v to zero increases selected reporting TVD to approximately 0.071 and 0.068, respectively. These interventions show sensitivity to the width coordinate but do not substitute for independently retrained width-blind models.

For the bounded estimator, write $R = \psi(z \oplus a)/\psi(z)$ and Δ for the learned phase difference. We replace each real moment sample $\mathrm{Re}[Re^{i\Delta}]$ with

$$\widetilde{Y} = \frac{2\,\mathrm{Re}[Re^{i\Delta}]}{1 + |R|^2}.$$

This is the reference-probability-weighted average over the unordered pair $\{z, z \oplus a\}$. It preserves the moment mean and satisfies $|\widetilde{Y}| \le 1$. Applying it to independent reference draws preserves the diagonal-excluded squared-moment estimator. Fixed-mask conditional averaging reduces moment and gradient variance, but shared draws across masks do not provide a general aggregate-variance guarantee. The selected reporting TVDs are 0.0289 ± 0.0057 and 0.0279 ± 0.0029, with mixed signs of the paired accuracy differences. These runs therefore do not establish a practical fidelity benefit from pair averaging.

The mean-response identity above also yields a direct finite-data calibration. For each fitting condition $c = (u_c, v_c)$, define $g_c = A(v_c/40)\sin(u_c/2)$ and let $\overline{K}_c$ be its fitting-pool mean. Least squares estimates

$$\widehat{s} = \frac{\sum_c g_c(\overline{K}_c - 8)}{\sum_c g_c^2}, \qquad \widehat{\theta} = \frac{\pi - \arcsin[\mathrm{clip}(\widehat{s}, -1, 1)]}{2}.$$

The previously chosen branch retains positive width response. This uses fitting observations and the encoder's response law, without validation data or population target probabilities. For the product prior, the same calculation uses $A(b) = 8\cos^{15}(b)$. Both controls give slightly lower mean TVD than the corresponding sampled fits (Table XI). They calibrate this particular one-angle model and are not general substitutes for sampled conditional training.

Checkpoint and selection replay, paired-input checks, and independent circuit comparisons support the reported finite-validation results. The bounded loss and gradient agree with the standalone estimator, and the original estimator reproduces the frozen short training trajectory. Efficient reference access, estimator control and output fidelity remain distinct properties. In particular, the finite-validation comparison does not establish an advantage from the bounded estimator or from Gaussian prior preparation.

Appendix C: Sixteen-qubit conditional switching construction

1. Graph geometry, cycle reference and symmetry

We order the sixteen edges lexicographically as

$$(01, 02, 03, 04, 05, 06, 12, 13, 14, 15, 16, 23, 24, 25, 26, 34). \tag{C1}$$

The chord set is $\mathcal{C} = \{12, 13, 14, 15, 16, 23, 24, 25, 26, 34\}$. Each chord and its two root edges form a fundamental triangle. We represent these ten triangles using a binary matrix $B \in \{0,1\}^{10\times 16}$. Each row contains ones on its triangle's three edges and zeros elsewhere. The chord columns form the identity matrix, so B has rank ten over $\mathbb{F}_2$. The connected graph has cycle rank $16 - 7 + 1 = 10$, so these rows form a complete cycle basis. The six independent vertex switches generate the set G of 64 switching masks. Each vertex switch flips either zero or two edges of every triangle, which gives $Bg = 0$ for every $g \in G$.

We denote the ten chord bits by $r \in \{0,1\}^{10}$ and write $|r|$ for their number of ones. Before the final Hadamard layer, we prepare the reference state

$$|\psi_\beta\rangle = \sum_{r\in\{0,1\}^{10}} a_\beta(r)\,|rB\rangle\,, \tag{C2}$$

$$a_\beta(r) = c_\beta^{10-|r|}(-is_\beta)^{|r|}, \tag{C3}$$

where $c_\beta = \cos(t_\beta/2)$, $s_\beta = \sin(t_\beta/2)$, and $t_\beta = (\pi/2)\tanh(1.5\beta)$. The product rB, evaluated modulo two, specifies the sixteen edge bits obtained by combining the triangles selected by r. Applying the Hadamard layer gives the encoded input $|\phi_\beta\rangle = H^{\otimes 16}\,|\psi_\beta\rangle$.

To sample the reference distribution classically, we draw the ten chord bits independently, assigning each a value of one with probability s_β^2, and return rB. We recover r directly from the chord bits because the corresponding columns of B form the identity matrix.

For a bit-flip mask in the cycle space, we write $m = uB$, with $u \in \{0,1\}^{10}$. Applying this mask maps rB to $(r \oplus u)B$. For nonzero β, we evaluate the amplitude ratio as

$$\frac{a_\beta(r\oplus u)}{a_\beta(r)} = \left[-i\tan\left(\frac{t_\beta}{2}\right)\right]^{|r\oplus u|-|r|}. \tag{C4}$$

If m lies outside the cycle space, the flipped configuration has zero reference amplitude, so the ratio is zero. At $\beta = 0$, the reference state is $|0\rangle^{\otimes 16}$: the ratio equals one for $m = 0$ and zero otherwise. We evaluate ratios only at sampled configurations with nonzero reference amplitude. These operations provide classical sampling and amplitude-ratio access without enumerating the full sixteen-qubit distribution.

Because $Bg = 0$, every supported configuration rB has even parity on the edges selected by the switching mask g. The corresponding Pauli-Z string therefore satisfies $Z^g\,|\psi_\beta\rangle = |\psi_\beta\rangle$. The Hadamard layer converts this relation into $X^g\,|\phi_\beta\rangle = |\phi_\beta\rangle$. Every IQP gate generated by a Pauli-X string commutes with these switching operators, so the learned circuit preserves switching symmetry for arbitrary parameter values.

We share one angle across the root edges and another across the chords to reduce the number of trainable parameters. Switching symmetry does not require this restriction. These two groups do not coincide with the edge orbits under graph-preserving vertex relabelings.

2. Exact quotient evaluation and validation

We group configurations related by vertex switches into classes labeled by $\ell = Bx$, evaluated modulo two. Each component of ℓ records the parity of the three edge bits in one fundamental triangle. Two configurations share a label exactly when vertex switches relate them, so each class contains 64 physical states. To choose a unique representative, we apply switches that set all six root-edge bits to zero. The chord bits then equal ℓ. Since vertex switches preserve the target energy, we evaluate it on this representative and denote it by $E(\ell)$.

We denote the total probability of the 64 states in class ℓ by $q_{\log}(\ell \mid \beta)$. Writing ϑ_e for the learned angle on edge e, we compute

$$q_{\log}(\ell \mid \beta) = \left|\frac{1}{32}\sum_r a_\beta(r) e^{i\sum_e \vartheta_e (-1)^{(rB)_e}} (-1)^{r\cdot\ell}\right|^2. \tag{C5}$$

In the two-angle model, we set $\vartheta_e = \alpha$ on root edges and $\vartheta_e = \gamma$ on chords. The sum over r includes all 2^{10} chord configurations, and $r\cdot\ell$ denotes their binary inner product. The factor $1/32 = 2^{-5}$ normalizes the ten-bit Hadamard transform.

We denote the target class probabilities by $p_{\log,\beta}(\ell)$. Both the learned and target distributions assign equal probability to all 64 states within each class. We therefore recover individual-state probabilities as

$$\begin{aligned} q(x \mid \beta) &= \frac{q_{\log}(Bx \mid \beta)}{64}, \\ p_\beta(x) &= \frac{p_{\log,\beta}(Bx)}{64}. \end{aligned} \tag{C6}$$

Each class contributes 64 identical terms to the full-distribution TVD, giving

$$\begin{aligned}&\frac{1}{2}\sum_{x\in\{0,1\}^{16}}|q(x\mid\beta)-p_\beta(x)|\\&\quad=\frac{1}{2}\sum_{\ell\in\{0,1\}^{10}}|q_{\log}(\ell\mid\beta)-p_{\log,\beta}(\ell)|.\end{aligned}\tag{C7}$$

This identity lets us evaluate the exact TVD of the full sixteen-qubit distribution using 2^{10} class probabilities.

We also evaluate switching-invariant observables exactly from the class probabilities, because each observable has the same value for all states in a class. Complete balance, $E = 16$, corresponds to the class $\ell = 0$, which contains 64 physical states. We therefore compute the learned probability of complete balance as $P_{\text{bal}} = q_{\log}(0 \mid \beta)$. This reduction enables exact classical evaluation of the benchmark; it does not establish a computational advantage.

We validate the encoder preparation and reduced class probabilities against direct sixteen-qubit state-vector calculations. We also compare sampled MMD estimates and their gradients with exact calculations using the kernel's Fourier representation, including the conversion from edge-angle gradients to gradients for the two shared angles. These checks test the circuit and estimator implementations. They do not establish that optimization finds the best parameters or that the model can represent the target distributions exactly.

3. Data, optimization and model selection

We train at $\beta \in \{0, 0.4, 0.8, 1.2\}$. For each fit, we draw 128 samples from the Gibbs target at each training temperature and keep the four sample pools fixed throughout optimization, giving 512 training samples in total. We select models using exact Gibbs distributions at the development temperatures $\beta \in \{0.18, 0.58, 0.98\}$. The sample count describes the training data; model selection also uses exact target information.

We evaluate the selected models at twelve additional temperatures:

$$\begin{aligned}\beta\in\{&0.03,0.11,0.27,0.37,0.53,0.67,\\&0.79,0.93,1.09,1.17,1.3,1.4\}.\end{aligned}\tag{C8}$$

We exclude these temperatures from training and model selection. We interpret the experiment as an exploratory assessment of conditional generalization.

At each update, we choose a training temperature uniformly and draw 256 target samples with replacement from its fixed sample pool. We also draw 512 fresh reference samples and 512 nonzero Fourier masks.

We use an equal mixture of two Gaussian Hamming kernels, with mask-bit probabilities 2/16 and 4/16 before excluding the zero mask. For a kernel band with bit probability p, we assign each sampled nonzero mask the weight $[1 - (1 - p)^{16}]/512$. The factor $1 - (1 - p)^{16}$ accounts for excluding the zero mask. We apply unbiased finite-sample corrections to the squared target moments and the squared model moments estimated from reference samples. We update the model parameters using sampled gradients and select checkpoints by exact mean TVD across the development temperatures.

The six candidate configurations cross initial learning rates $0.005, 0.015, 0.04$ with Gaussian initialization widths $0.1, 0.5$. Adam uses $(b_1, b_2) = (0.9, 0.999)$, $\epsilon = 10^{-8}$ and no weight decay. Cosine decay ends at one tenth of the initial learning rate. Six screening fits use 512 updates and checkpoints every 64 updates. The best two candidates each receive two independent 2000-update confirmation fits, with checkpoints every 100 updates. We select the configuration by mean confirmation development TVD, then freeze it before any final reporting evaluation. The selected learning rate is 0.015 with initialization width 0.5. Five independent final fits use 2000 updates. The initial and final checkpoints remain eligible in every stage.

4. Parameter sharing, controls and response diagnostics

An exact-distribution diagnostic at the development temperatures compares one common local angle, root/chord tying with two angles, and seven angles tied to edge orbits under independent exchanges of vertices 1/2, 3/4 and 5/6. The seven groups are $\{01, 02\}$, $\{03, 04\}$, $\{05, 06\}$, $\{12\}$, $\{13, 14, 23, 24\}$, $\{15, 16, 25, 26\}$ and $\{34\}$. Sixteen optimization starts give best observed development TVD values 0.52919, 0.18849 and 0.18322, respectively. These diagnostics fit exact distributions and do not constitute finite-data training results or global optimality certificates. We retain two angles because the observed gain from five additional parameters is 0.00527. The encoder itself uses one fixed temperature map and a fundamental cycle basis, with no extra preparation layers or ancillary qubits.

The encoder-only model is uniform, so its mean balance is zero and its fully balanced probability is 1/1024. Holding each selected model's input at $\beta = 0.6$ yields a sensitivity control, without retraining or selecting a temperature-blind comparator. A separate fixed classical control draws independent edge signs of mean $a_\beta = \sin t_\beta$ and then switches uniformly:

$$q_\beta^{\mathrm{fix}}(x) = \frac{1}{64} \sum_{g \in G} \prod_e \frac{1 + a_\beta (-1)^{x_e \oplus g_e}}{2}. \tag{C9}$$

It has exact symmetry and original-grid mean TVD 0.1134, compared with native TVD 0.1780 ± 0.0013. It limits performance claims but does not change the construction's exact symmetry guarantee.

We compute every observable within each fit, then report the mean and sample standard deviation across five fits. For the Hamiltonian $H = -E$, the per-edge fluctuation statistic is

$$c_{\mathrm{fluc},q}(\beta) = \frac{\beta^2}{16} \left(\langle E^2 \rangle_q - \langle E \rangle_q^2 \right). \tag{C10}$$

It equals equilibrium heat capacity for the Gibbs target; a generated distribution need not satisfy the fluctuation-response relation. At $\beta = 2.4$, the target gives 0.0024 and the native model gives 3.1929 ± 0.0528. Figure 11 displays this discrepancy. We compute the variance within each fit before aggregation, rather than from a mixture of fitted models.

A post-selection diagnostic evaluates the frozen models at 121 points over $\beta \in [0, 2.4]$ using triangle balance and fully balanced probability. We select neither hyperparameters nor checkpoints with these diagnostics. The dense grid includes fitting and development temperatures. Ten further temperatures, $1.5, 1.6, \ldots, 2.4$, provide a separate post-selection extrapolation summary; we keep it distinct from the original twelve-condition report. Training ends at 1.2, so the displayed interpolation and extrapolation widths are both 1.2. Agreement in the selected order observables does not imply fidelity of all correlations or thermodynamic response.

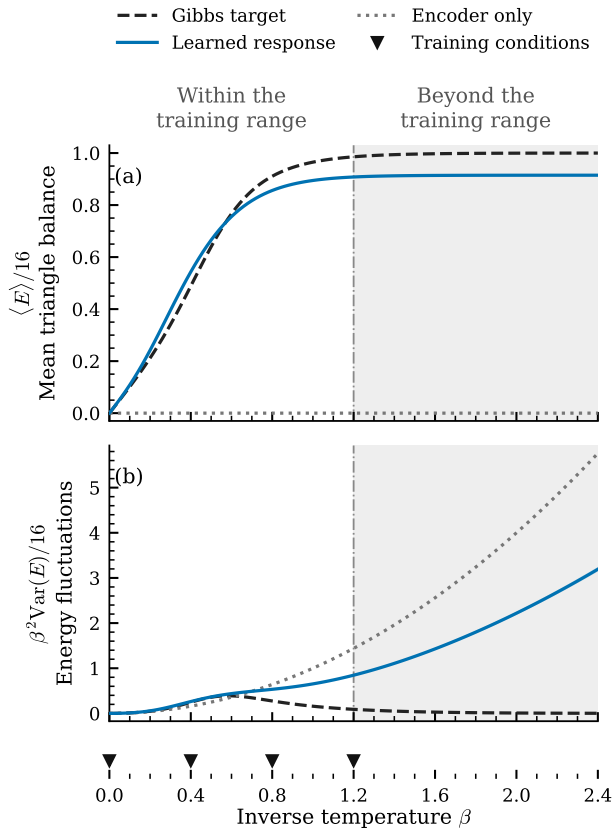


FIG. 11. Energy-fluctuation response of the learned distributions. (a) Mean triangle balance, repeated for context. (b) The fluctuation statistic $\beta^2 \mathrm{Var}(E)/16$. This statistic equals heat capacity per edge for the equilibrium Gibbs target; the generated distribution need not satisfy the fluctuation-response relation. Styles, sample-standard-deviation bands, fitting markers and shading follow Figure 7. The generated fluctuations remain substantially larger than the target at large inverse temperature. We compute the variance within each fit before aggregation.